FAKULTA APLIKOVANÝCH VĚD
ZÁPADOČESKÉ UNIVERZITY
V PLZNI

Edgebreaker

Komprese trojúhelníkových sítí

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1. Motivace

1.1 Dobový kontext

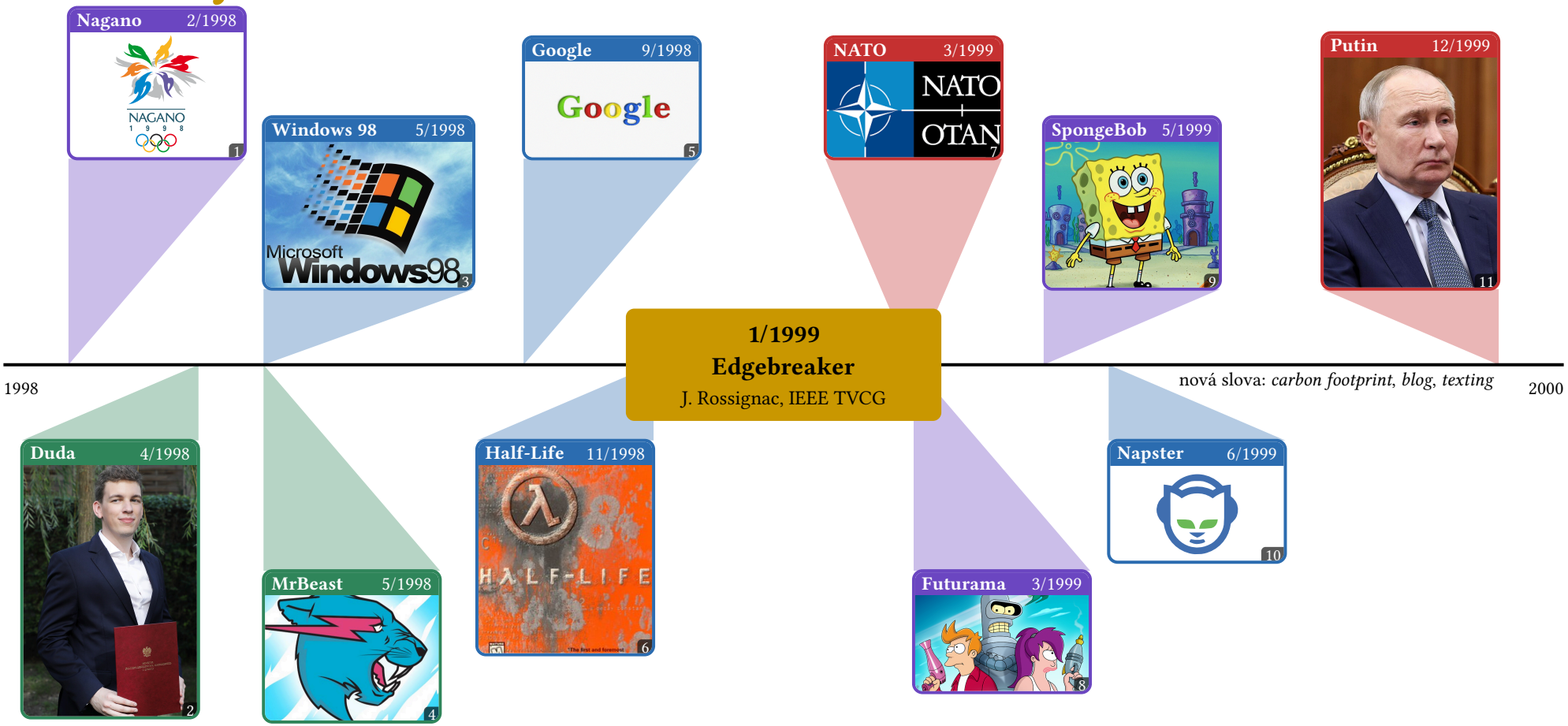


Figure 1: Dobový kontext vzniku algoritmu Edgebreaker.

1.1 Dobový kontext

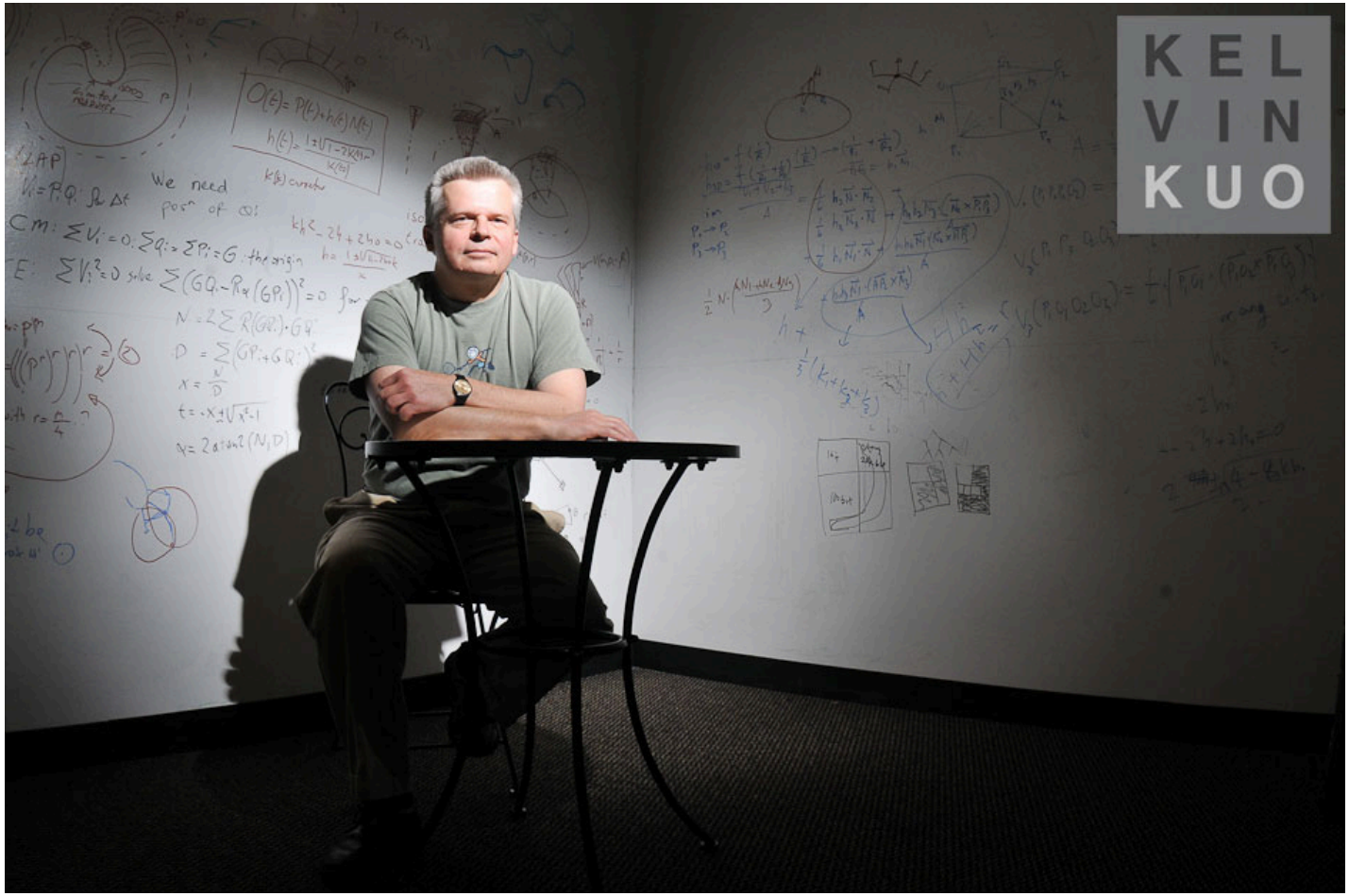


Figure 2: Jarek Rossignac

1.2 Dělení

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z)$$

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.2.2 Konektivita

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.2.2 Konektivita

$$\forall e \in E : e = \left(e_i, e_j\right)$$

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.2.2 Konektivita

$$\forall e \in E : e = \left(e_i, e_j\right)$$

$$\forall t \in T : t = \left(v_i, v_j, v_k\right)$$

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.2.2 Konektivita

$$\forall e \in E : e = \left(e_i, e_j\right)$$

$$\forall t \in T : t = \left(v_i, v_j, v_k\right)$$

$$\forall c \in 3T : c = \left(c_{\mathrm{op}}\right)$$

1.2 Dělení

1.2.1 Geometrie

$$\forall v \in V :$$

$$v = (x, y, z, n, \mathrm{uv}, \ldots)$$

1.2.2 Konektivita

$$V : 2F : 3E$$

2. Komprese

2.1 Pravidla

History =

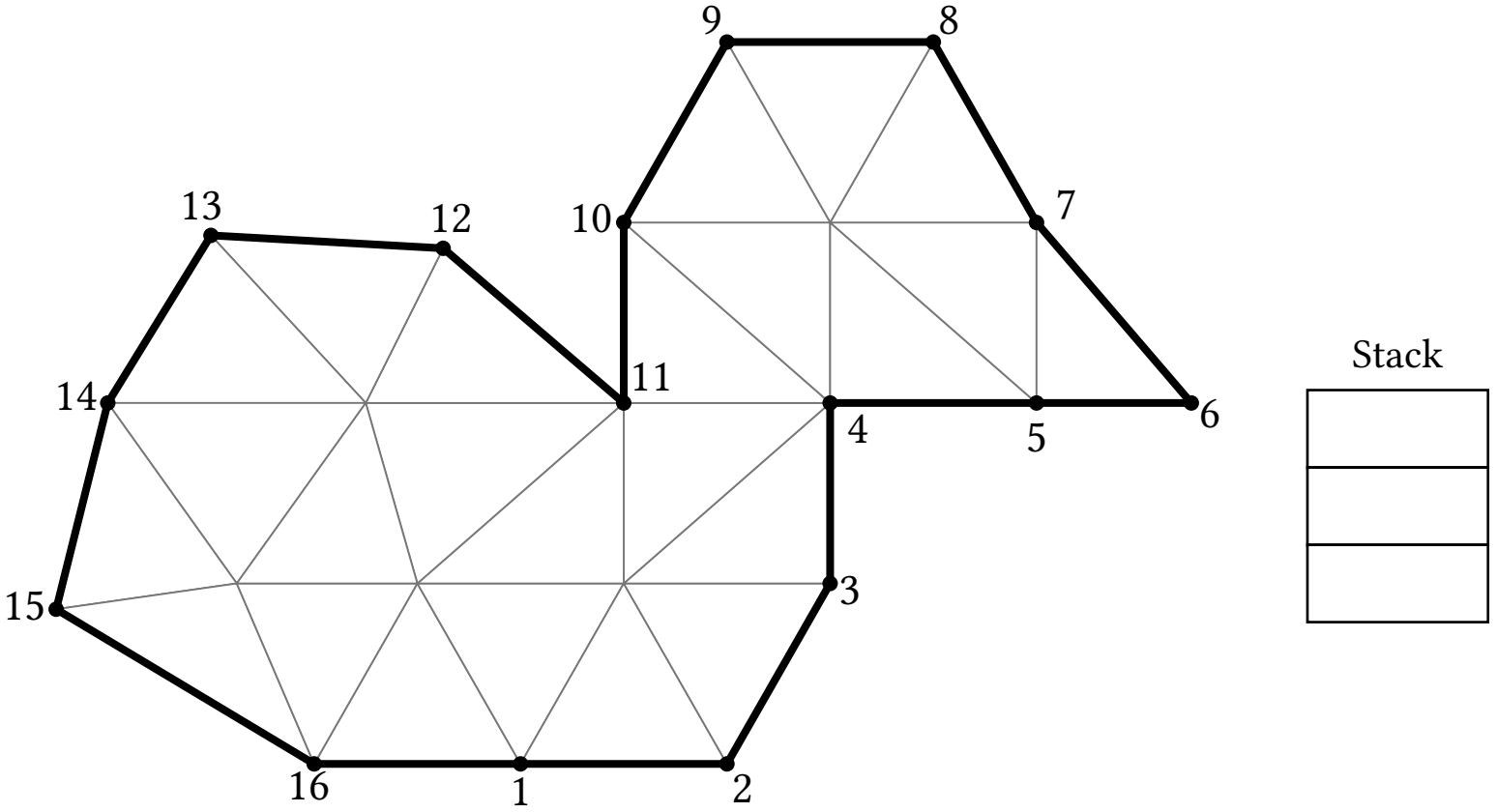


Figure 3: Jednoduchá komprese.

History = C

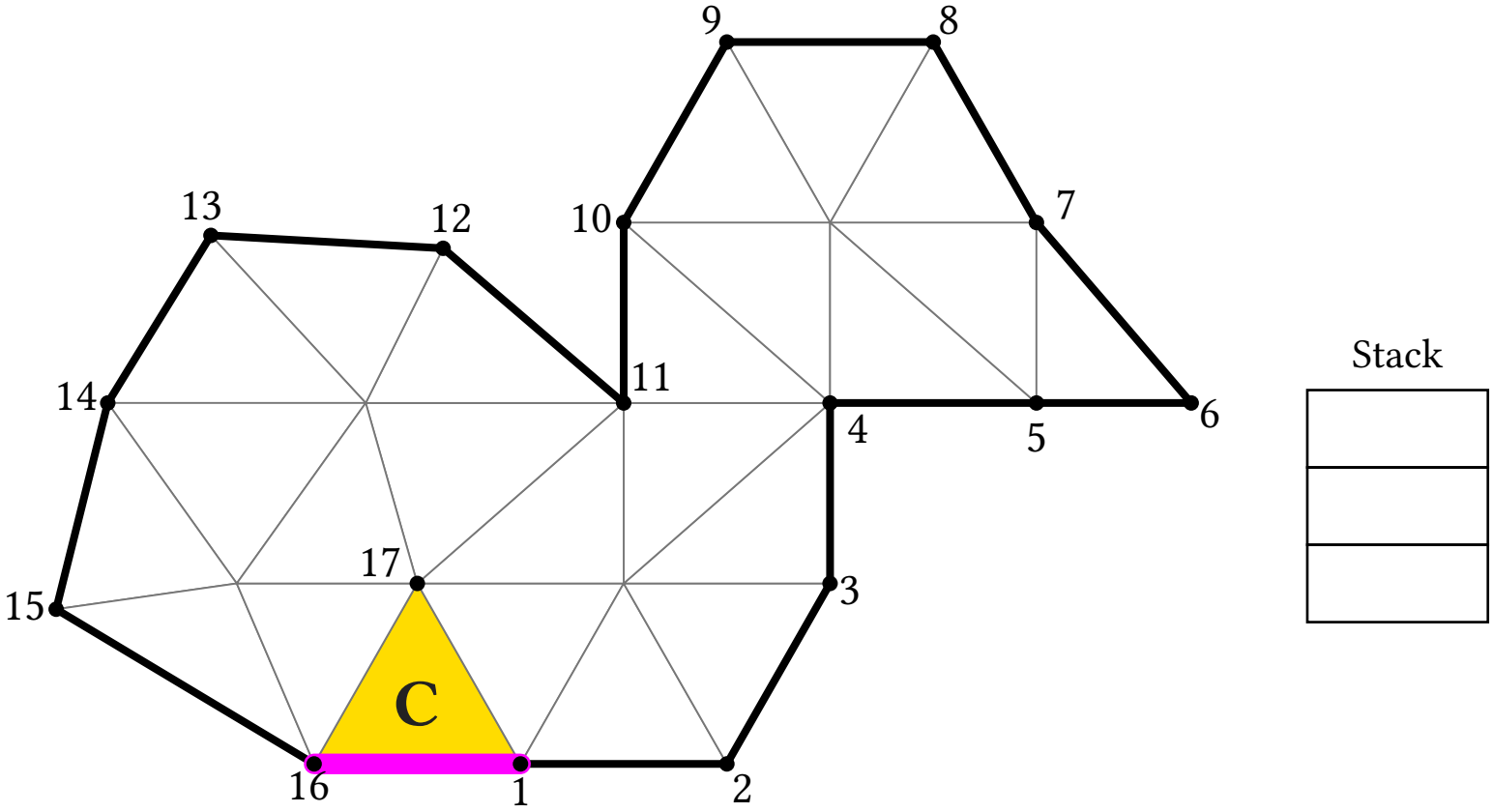


Figure 3: Jednoduchá komprese.

History = CC

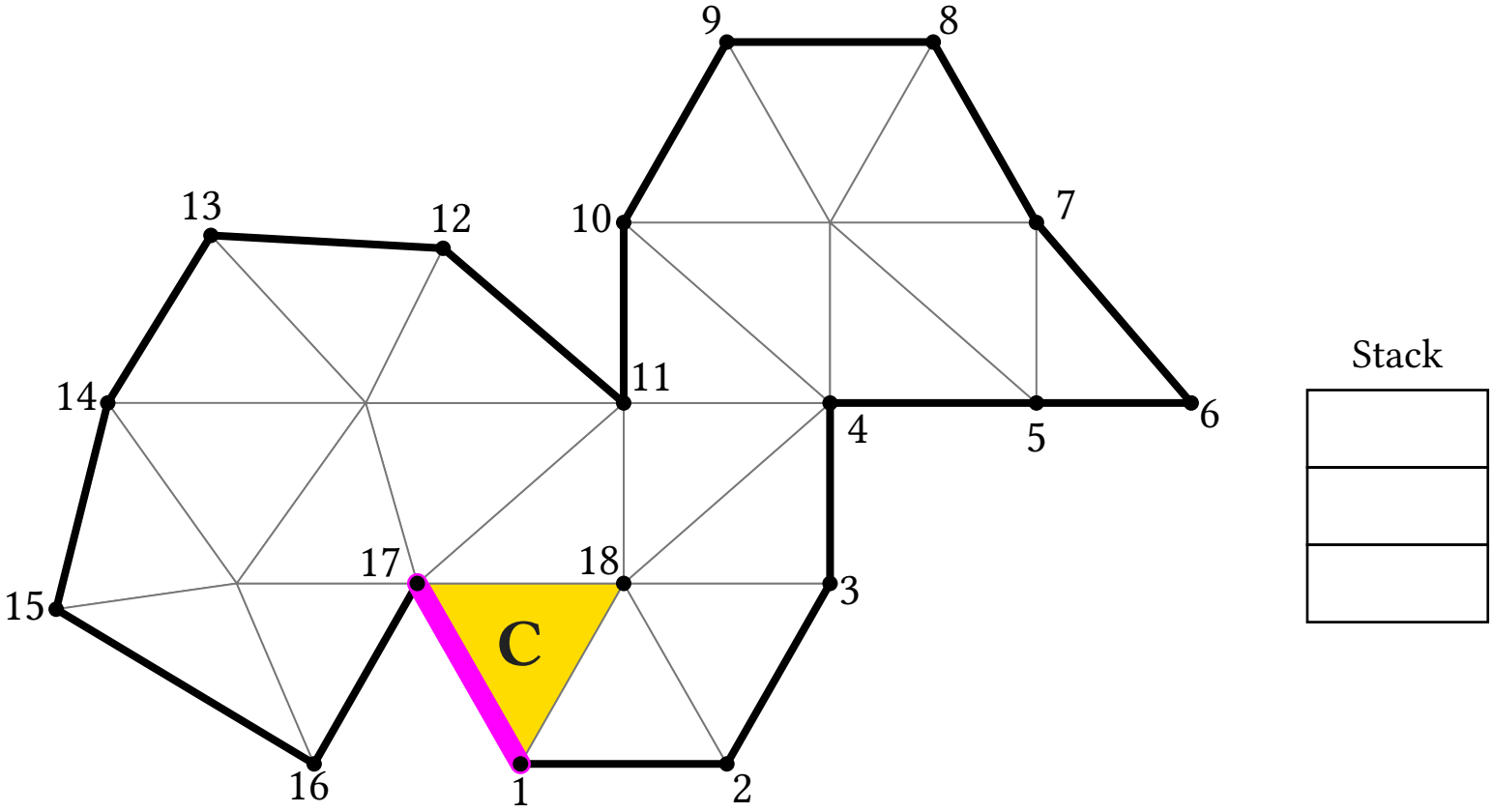


Figure 3: Jednoduchá komprese.

History = CCR

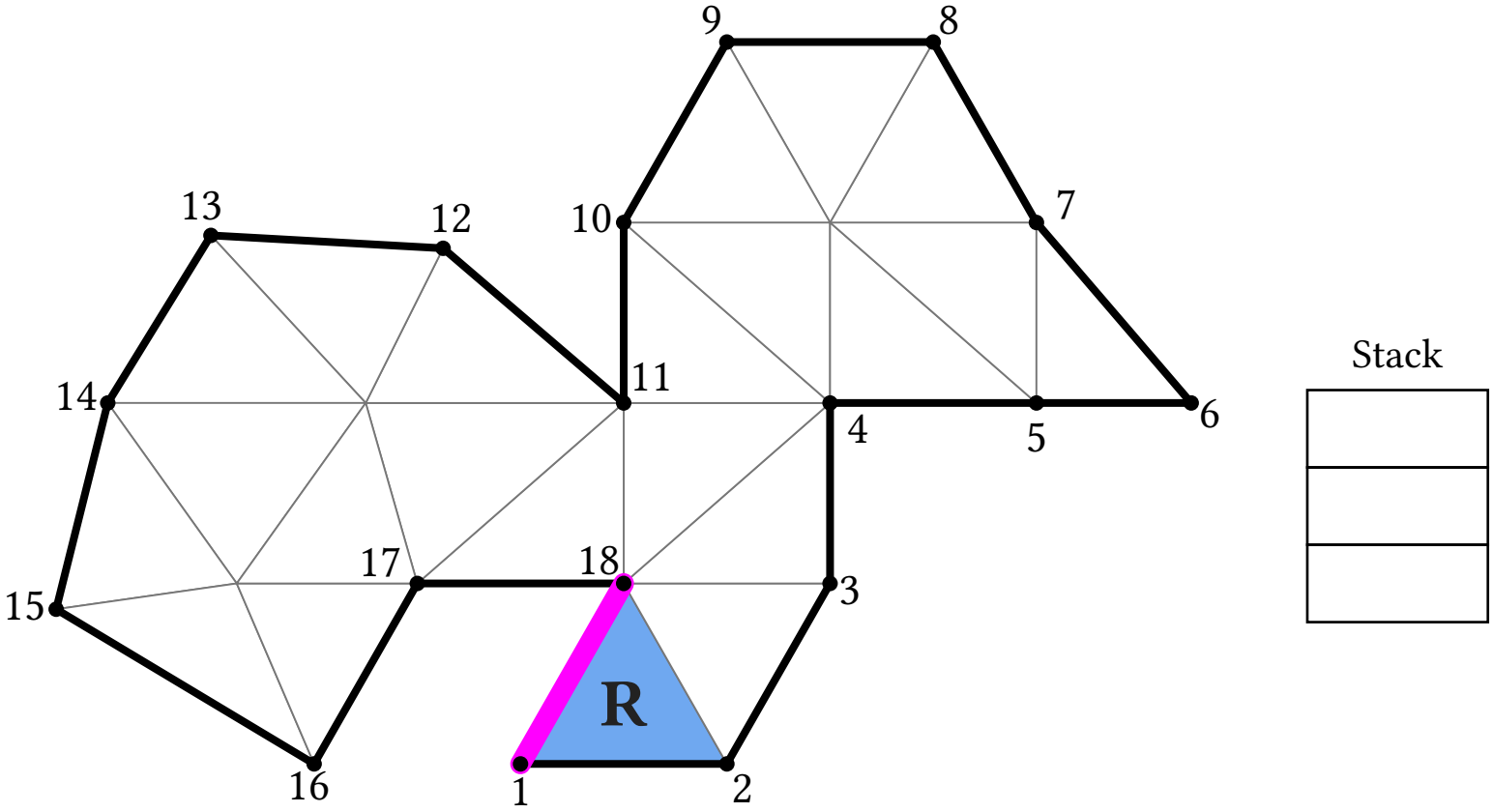


Figure 3: Jednoduchá komprese.

History = CCRR

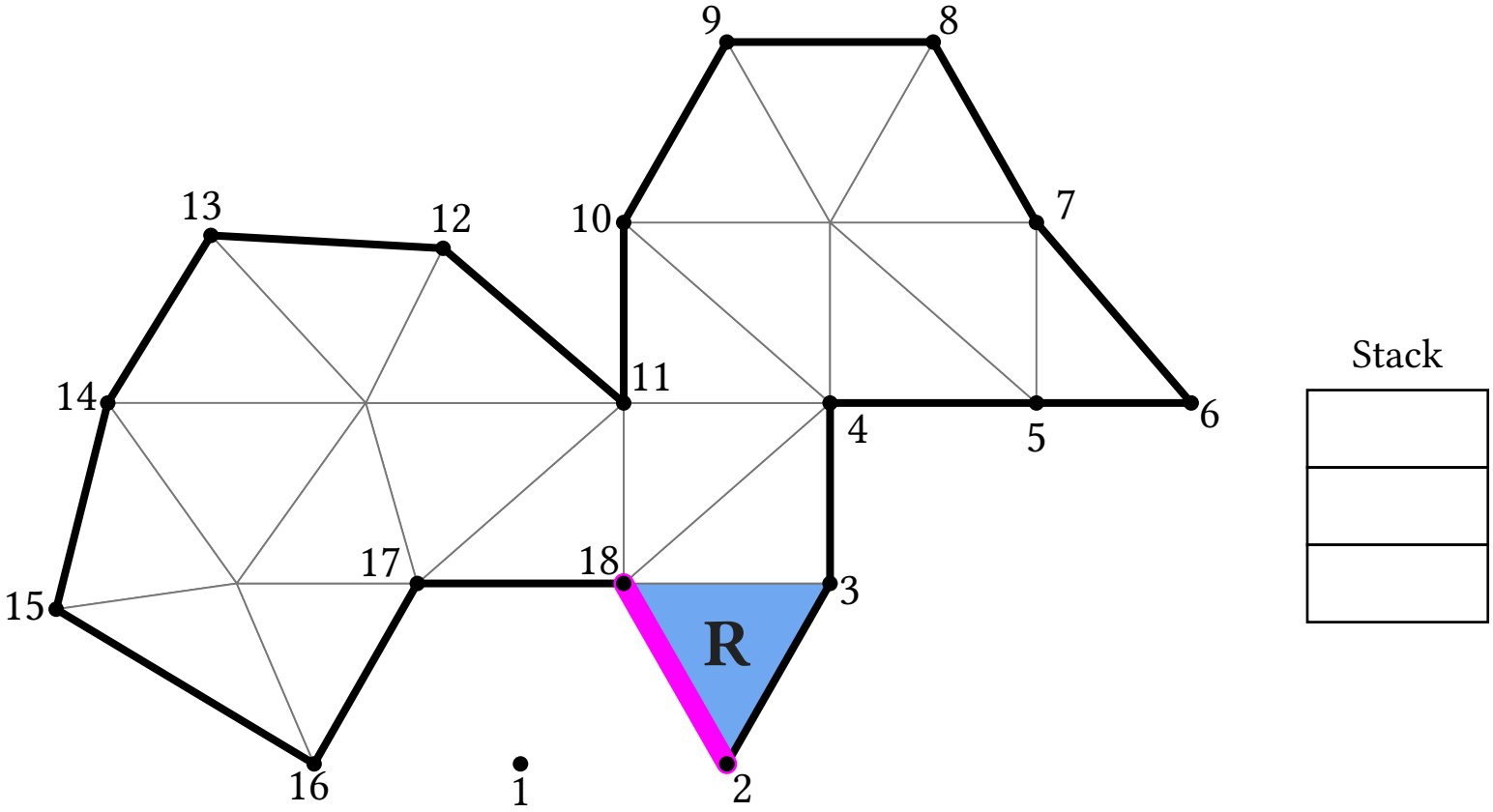


Figure 3: Jednoduchá komprese.

History = CCRRR

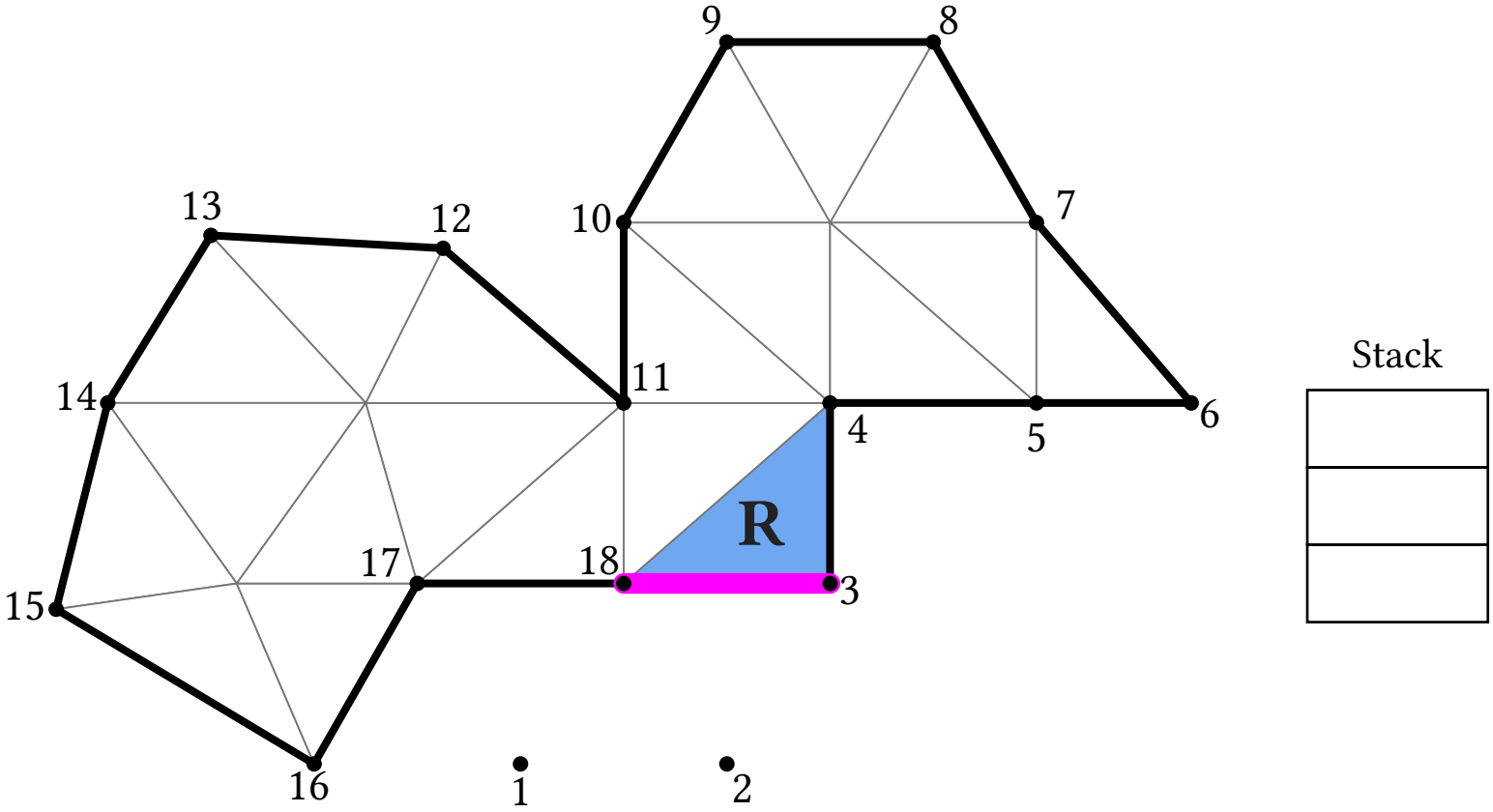


Figure 3: Jednoduchá komprese.

History = CCRRRS

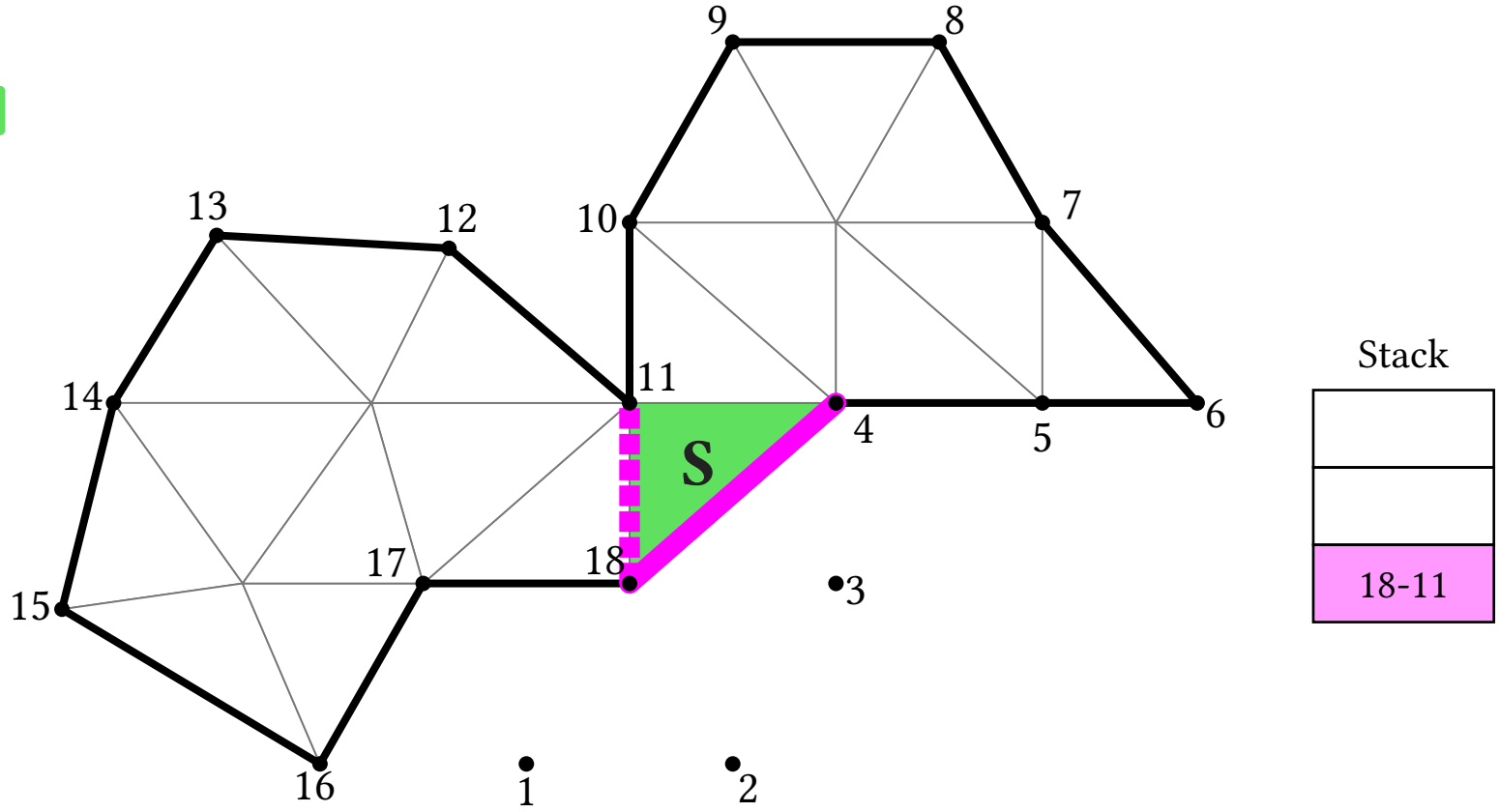


Figure 3: Jednoduchá komprese.

History = CCRRRSL

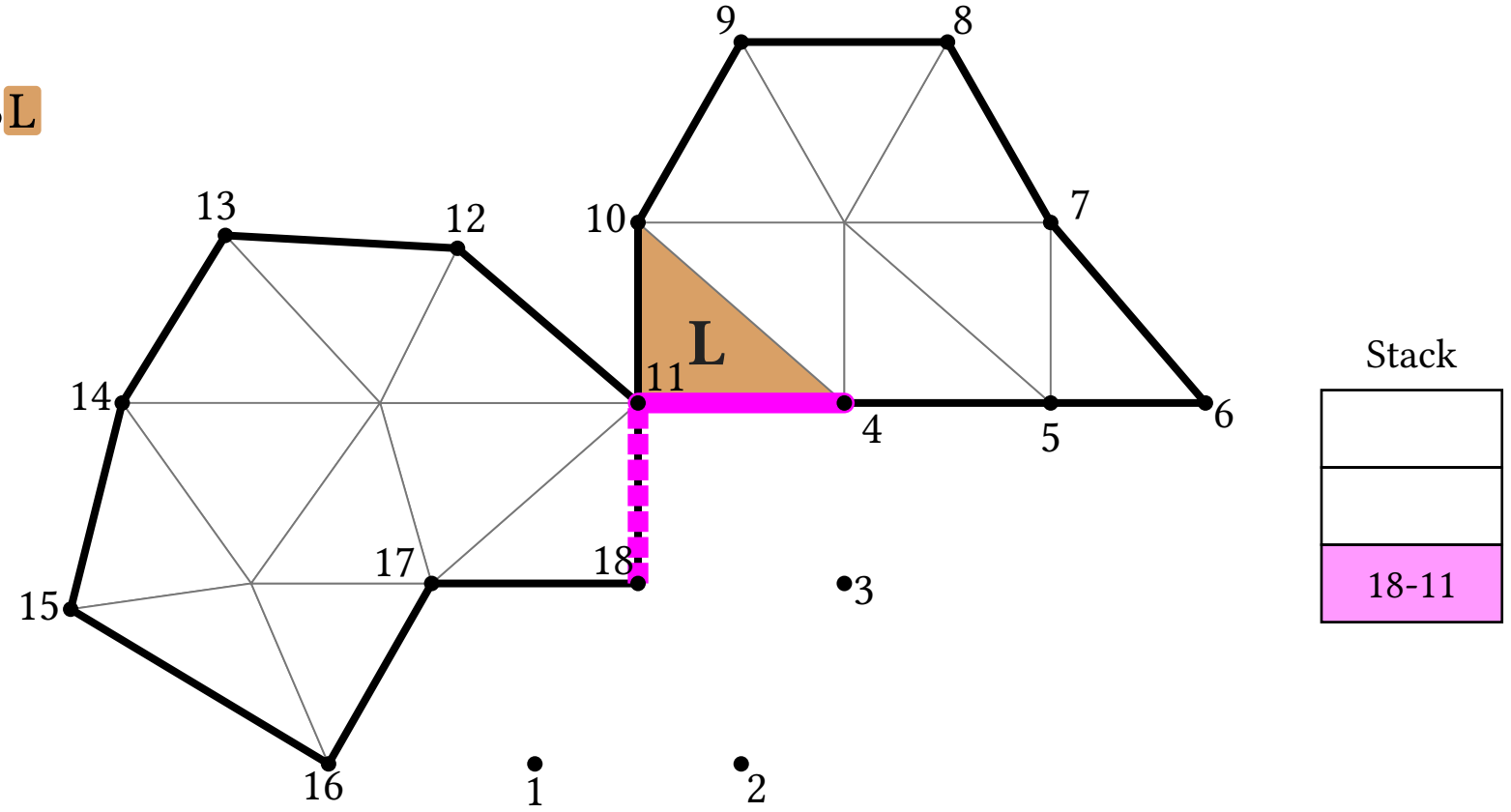


Figure 3: Jednoduchá komprese.

History = CCRRRSLC

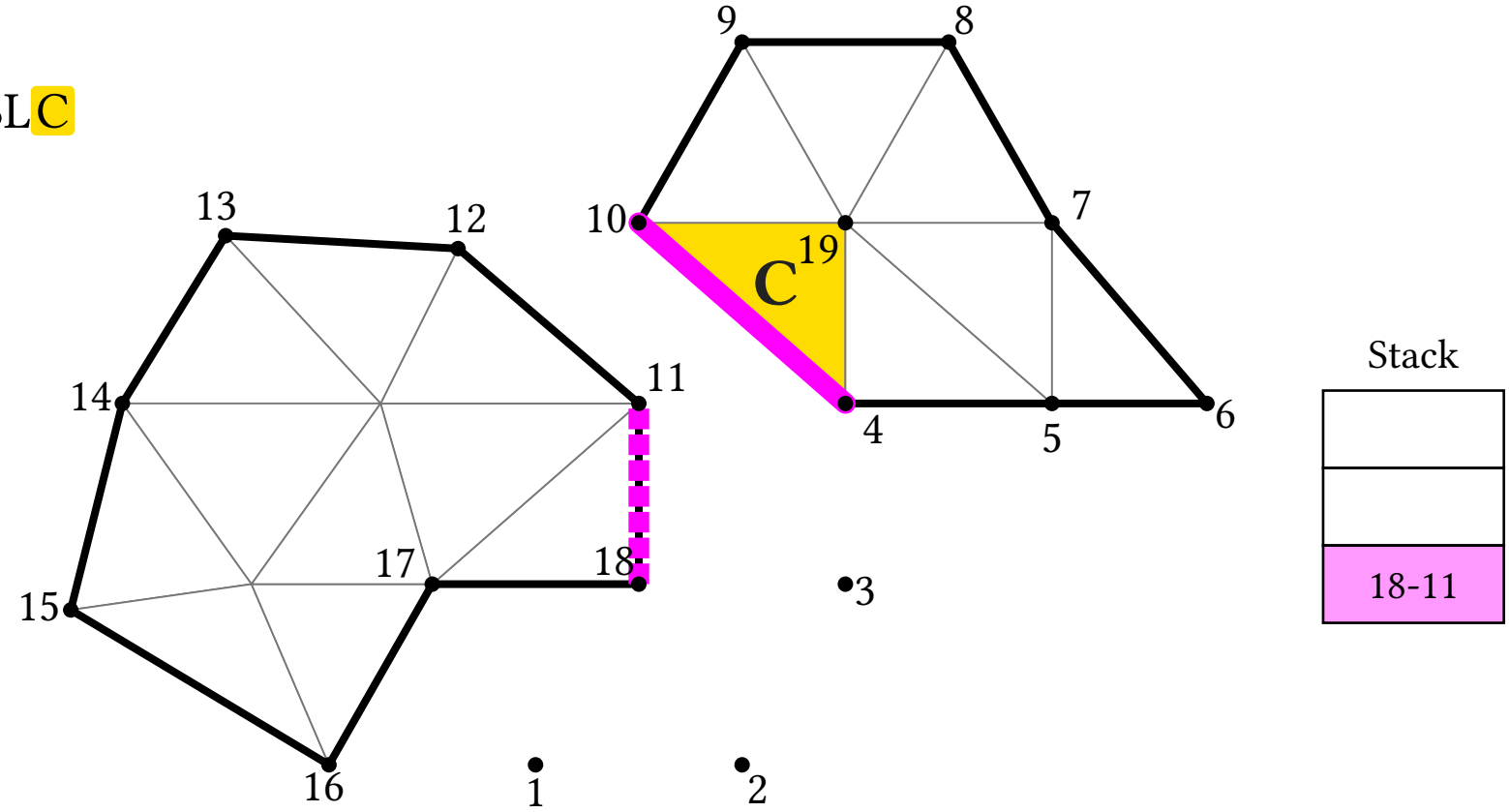


Figure 3: Jednoduchá komprese.

History = CCRRRSLCR

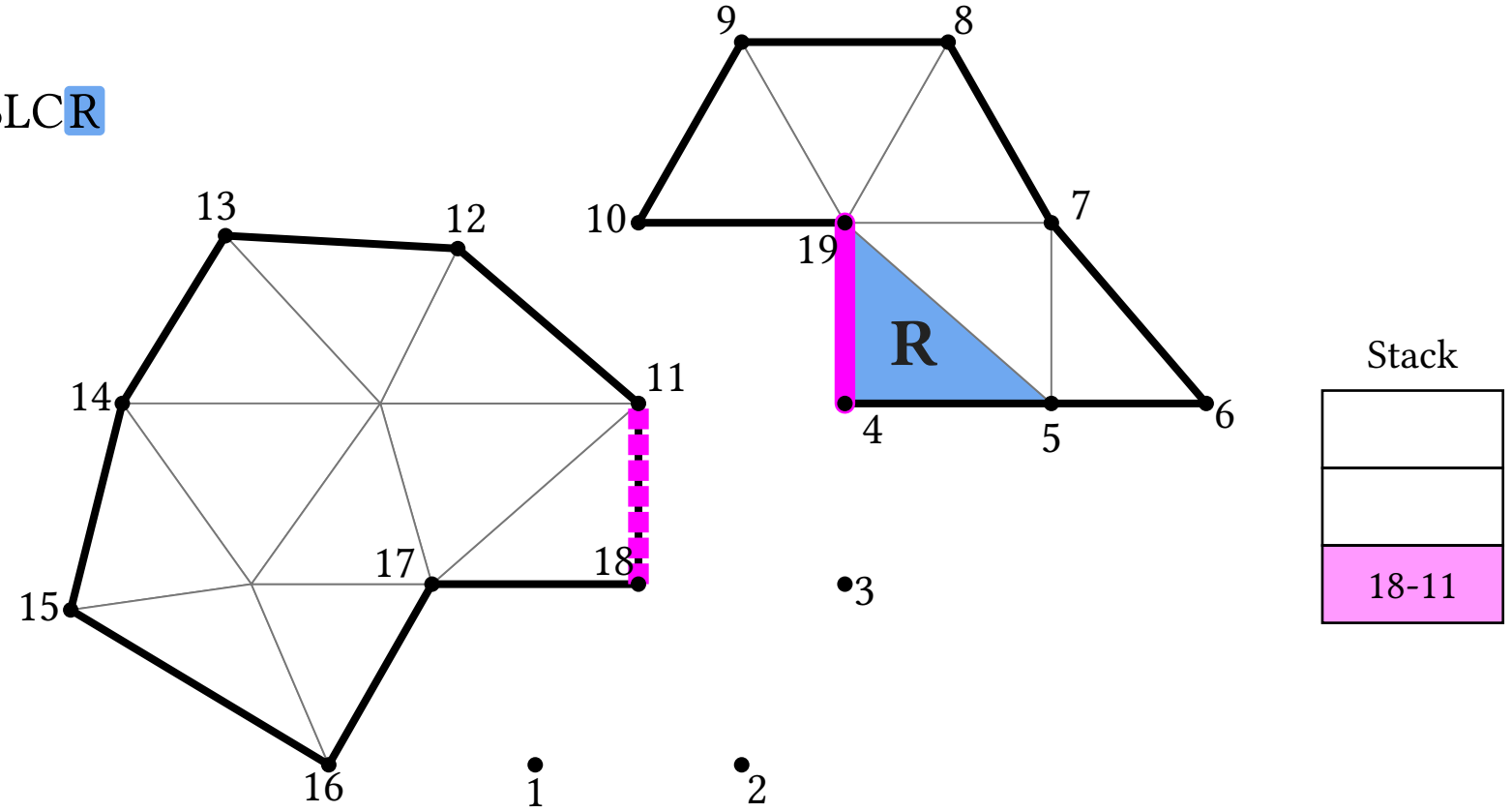


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRS

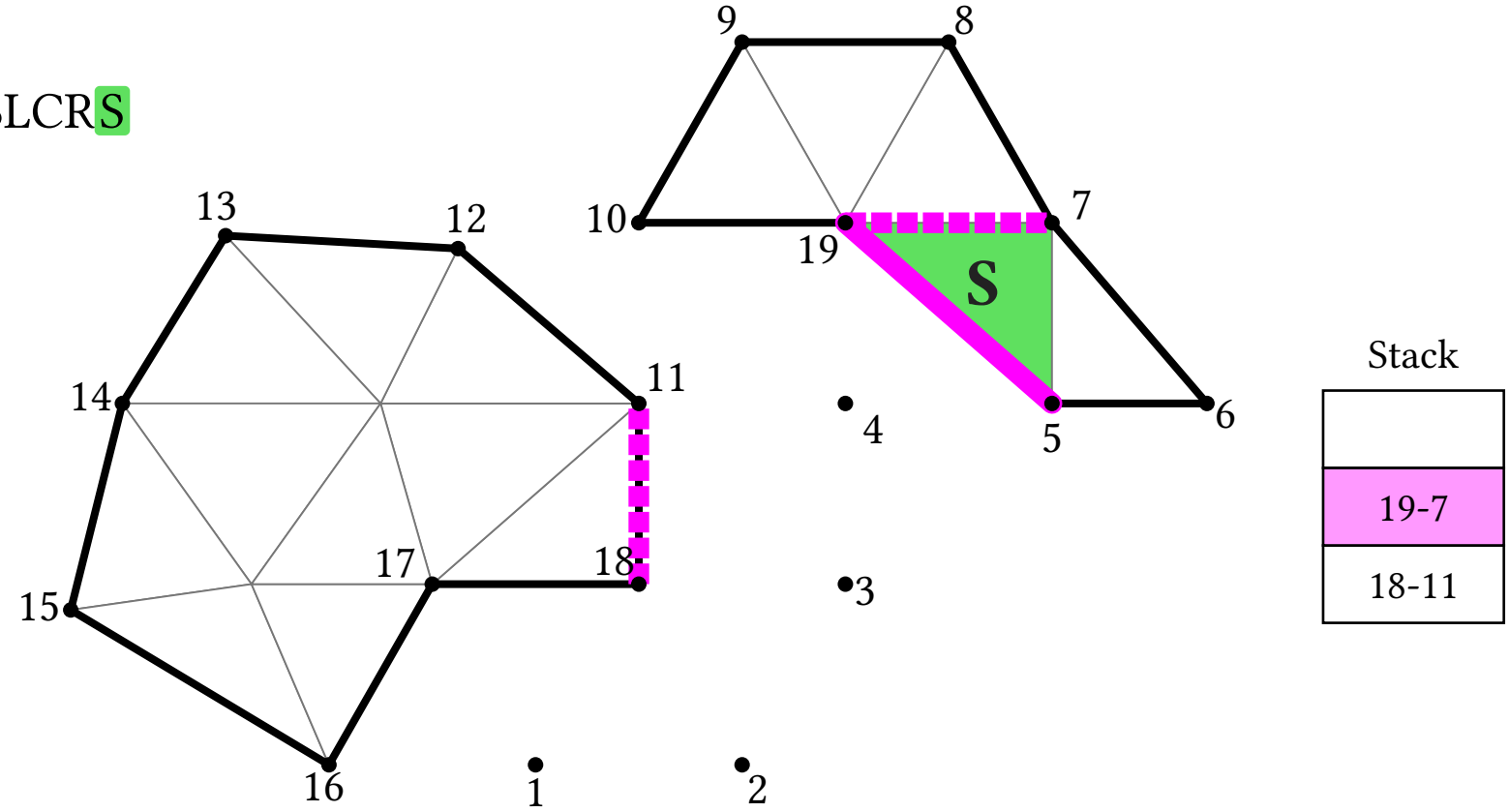


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSE

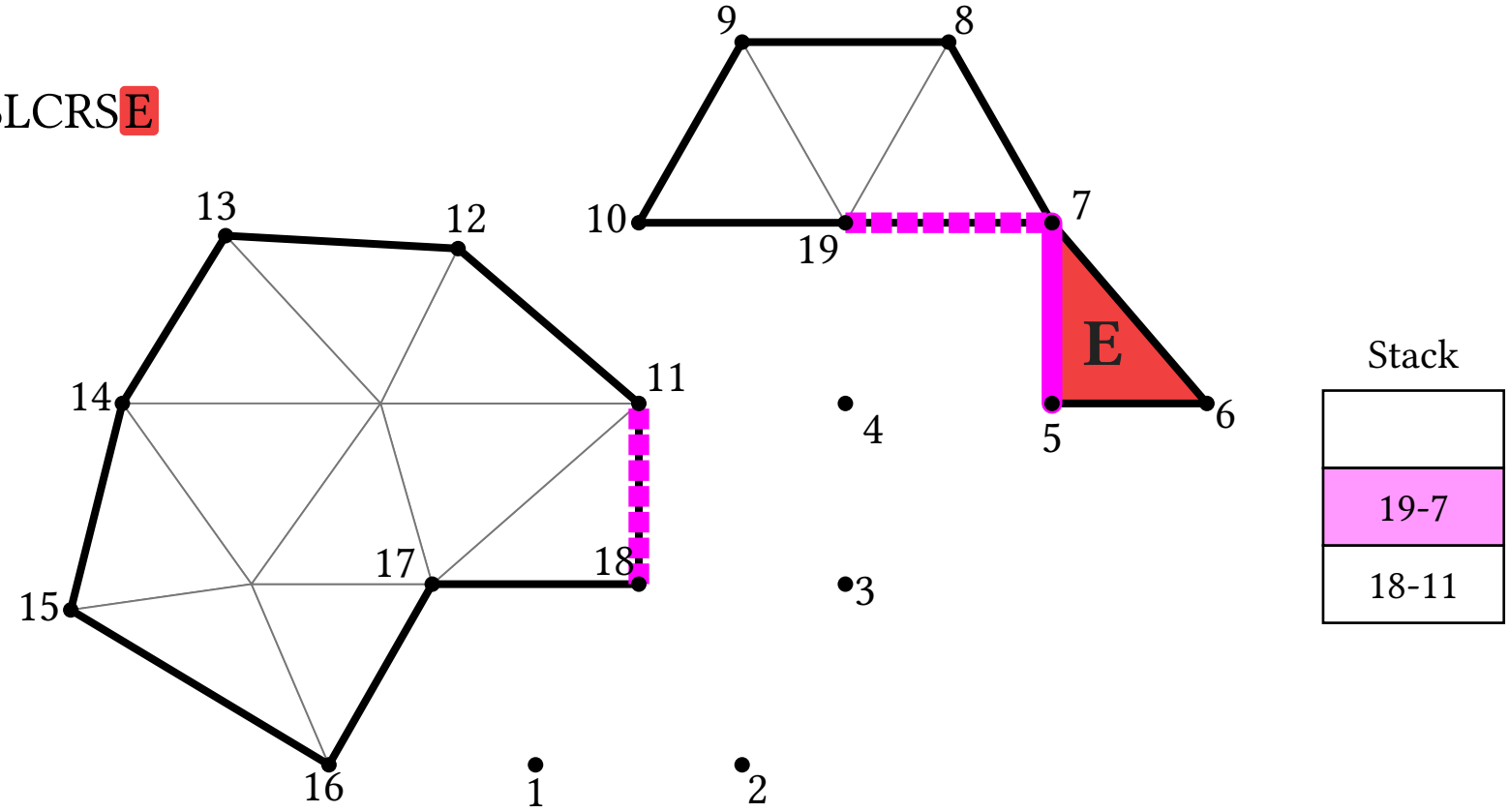


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSER

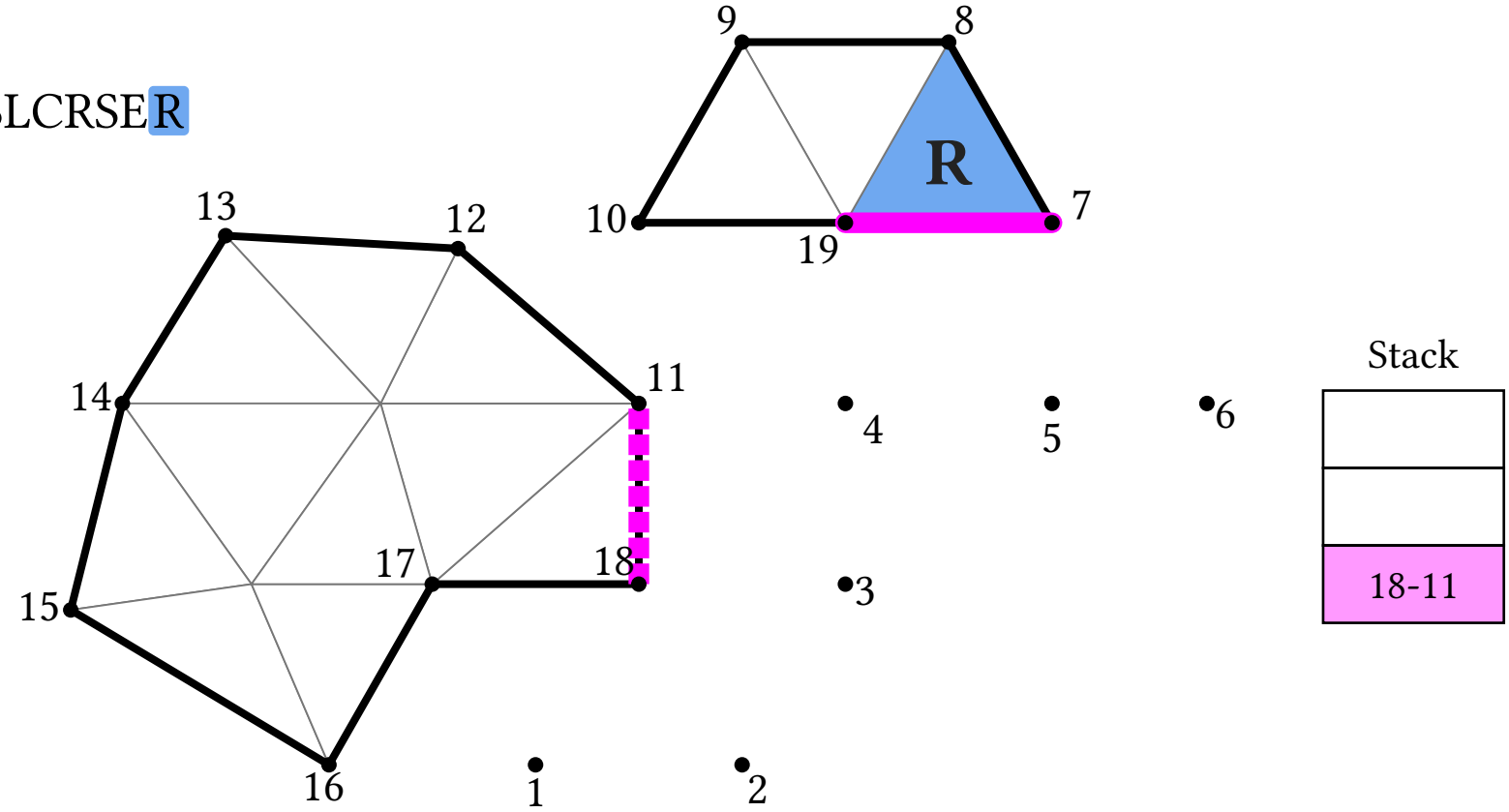


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERR

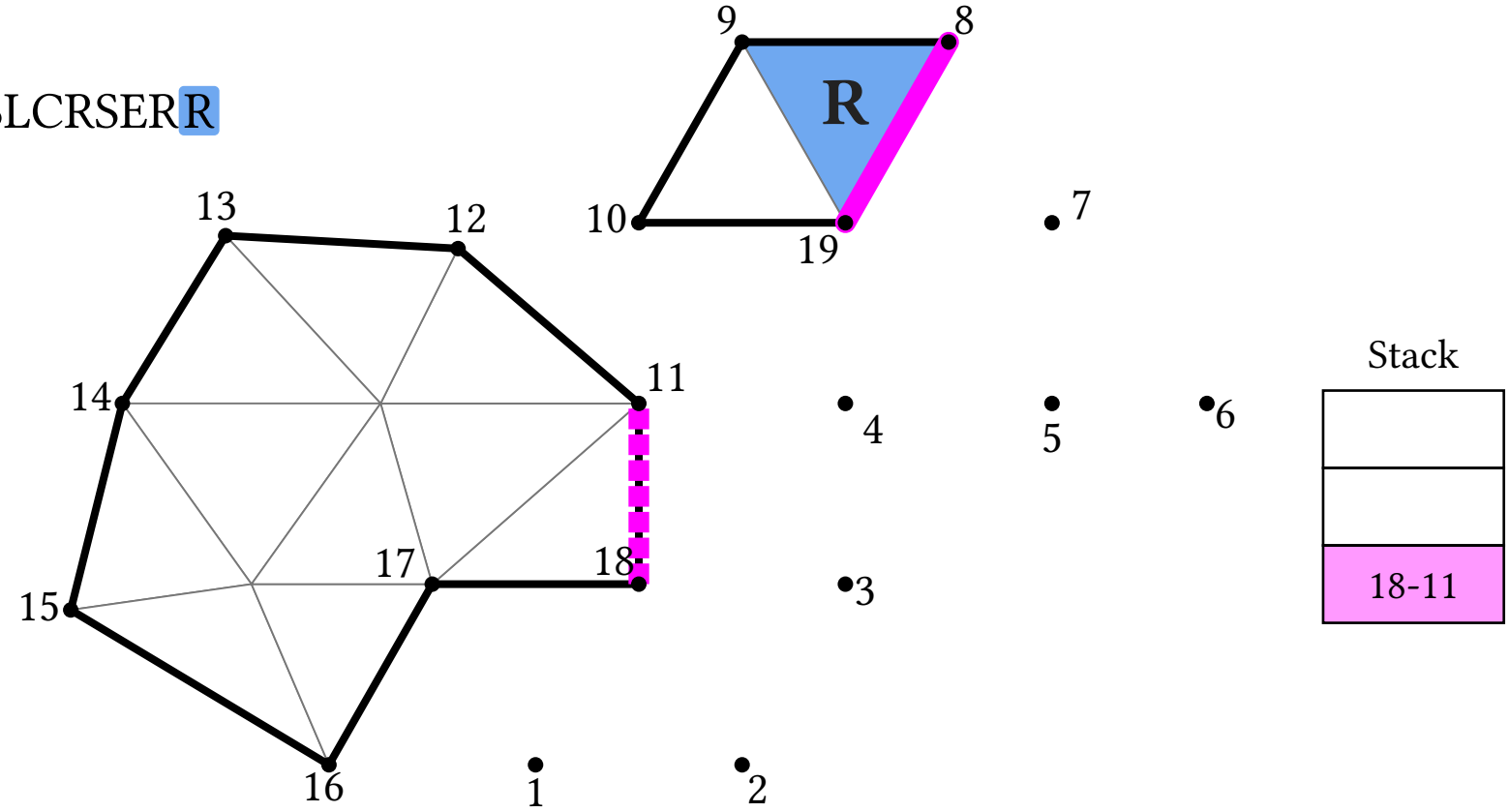


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRE

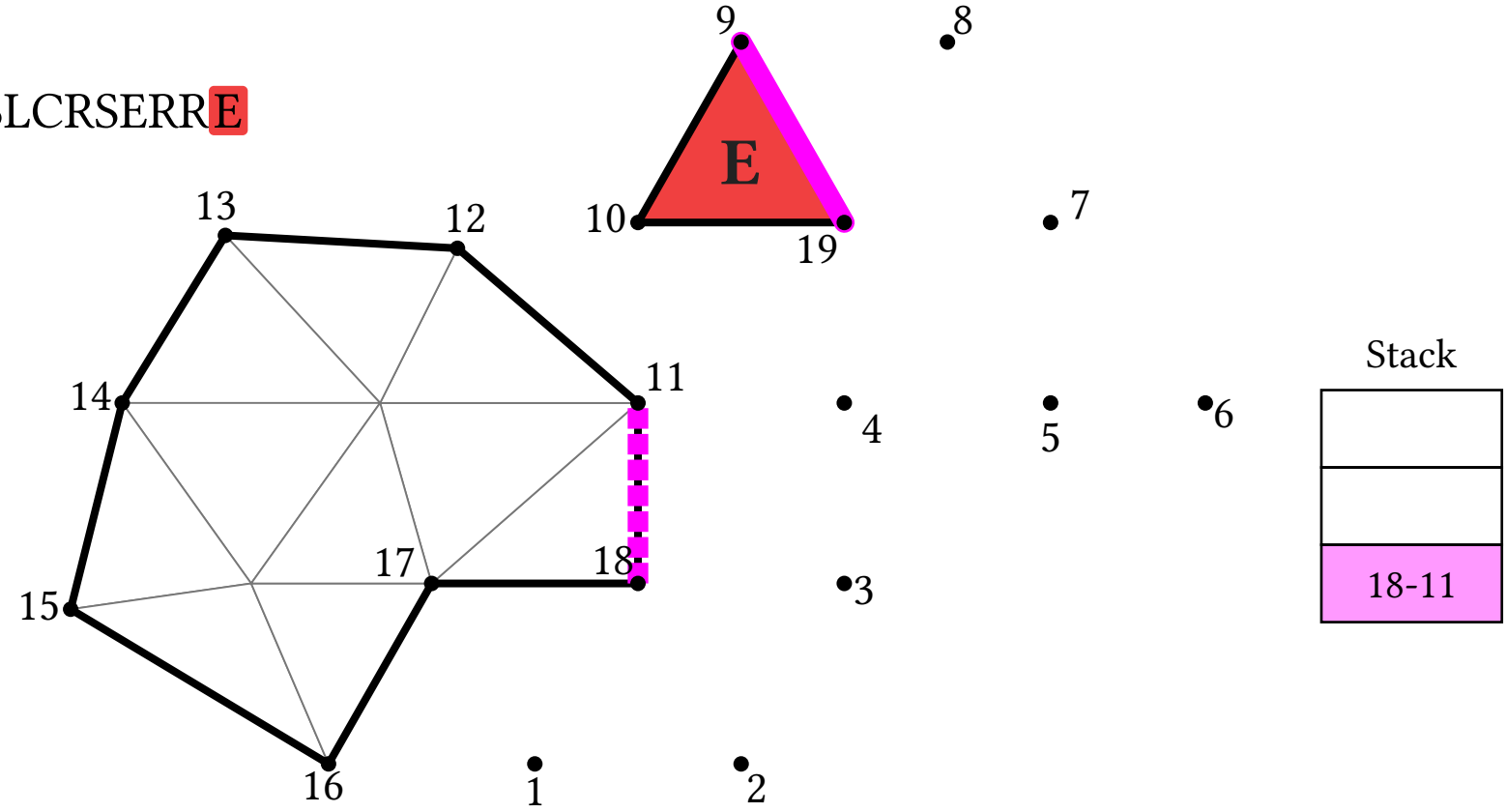


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERREL

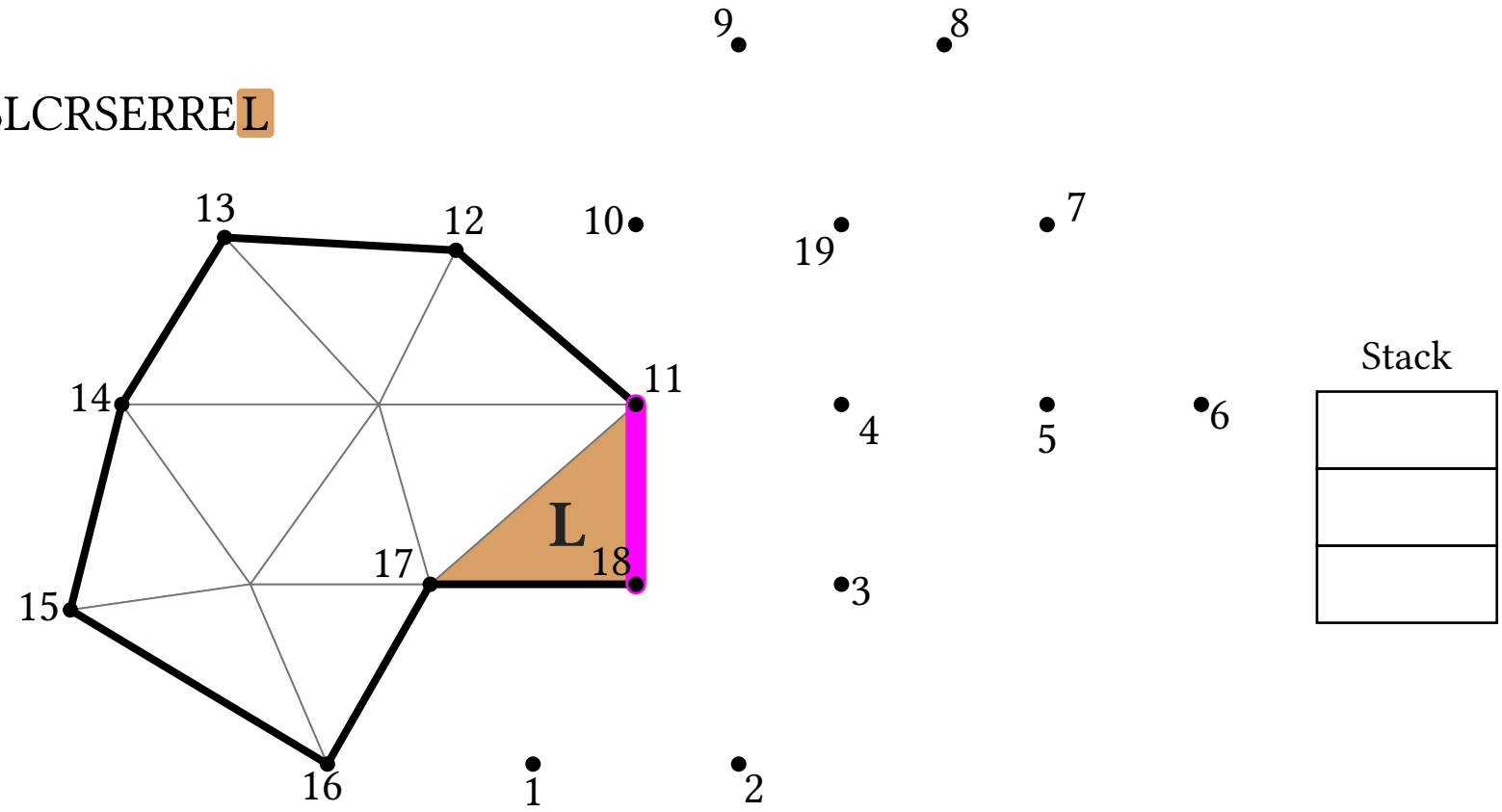


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERREL**C**

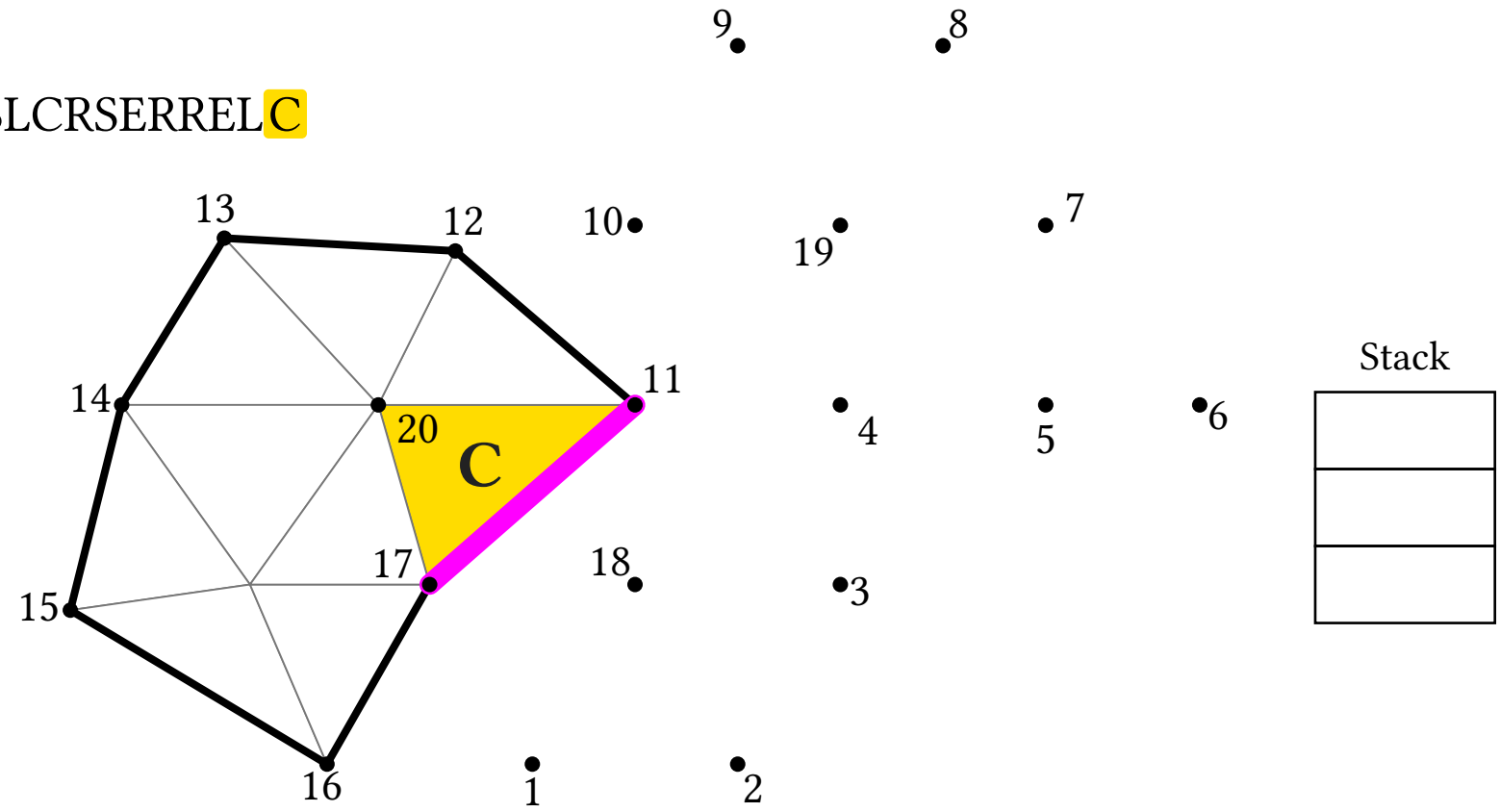


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCR

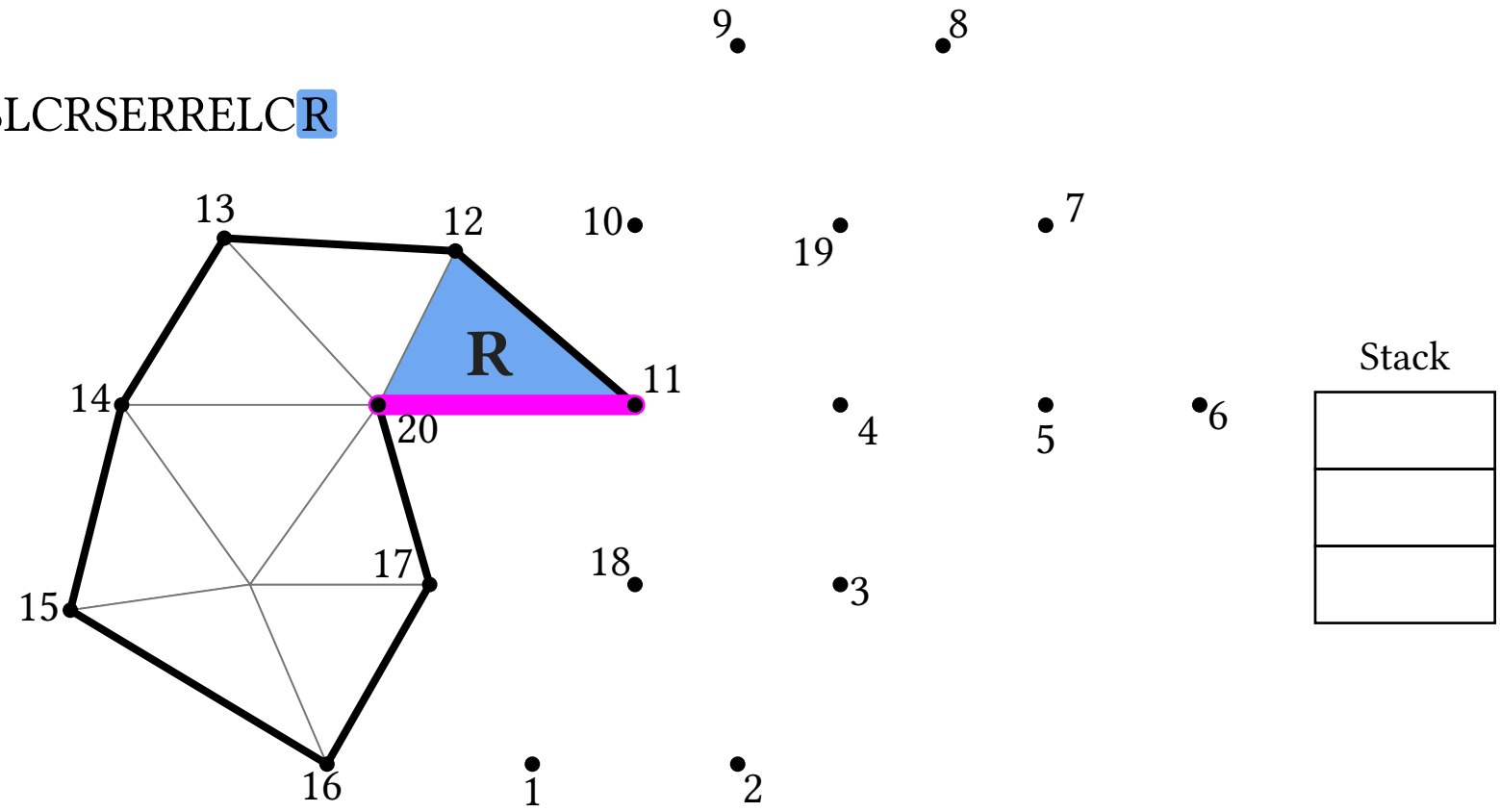


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRR

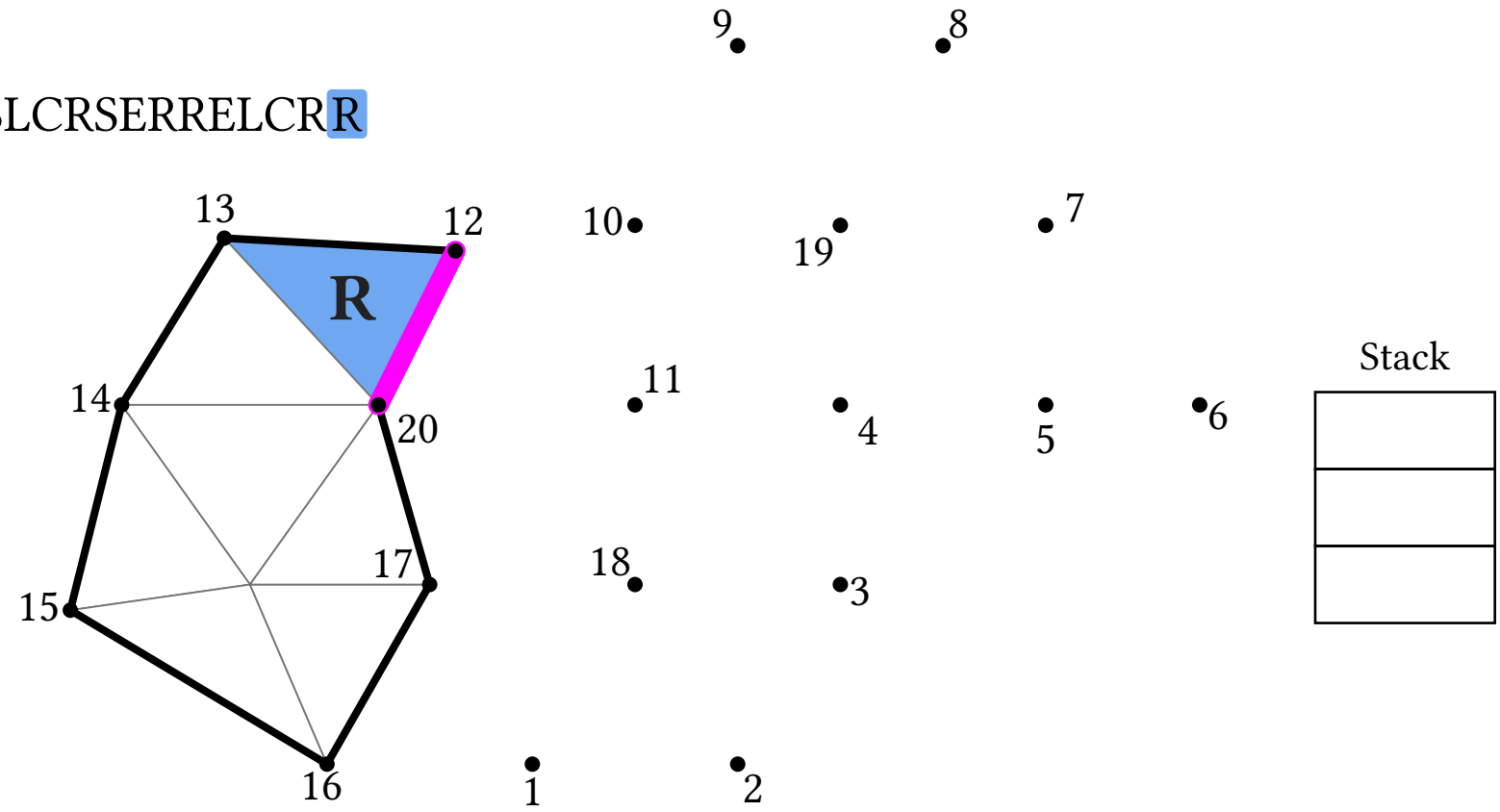


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRR

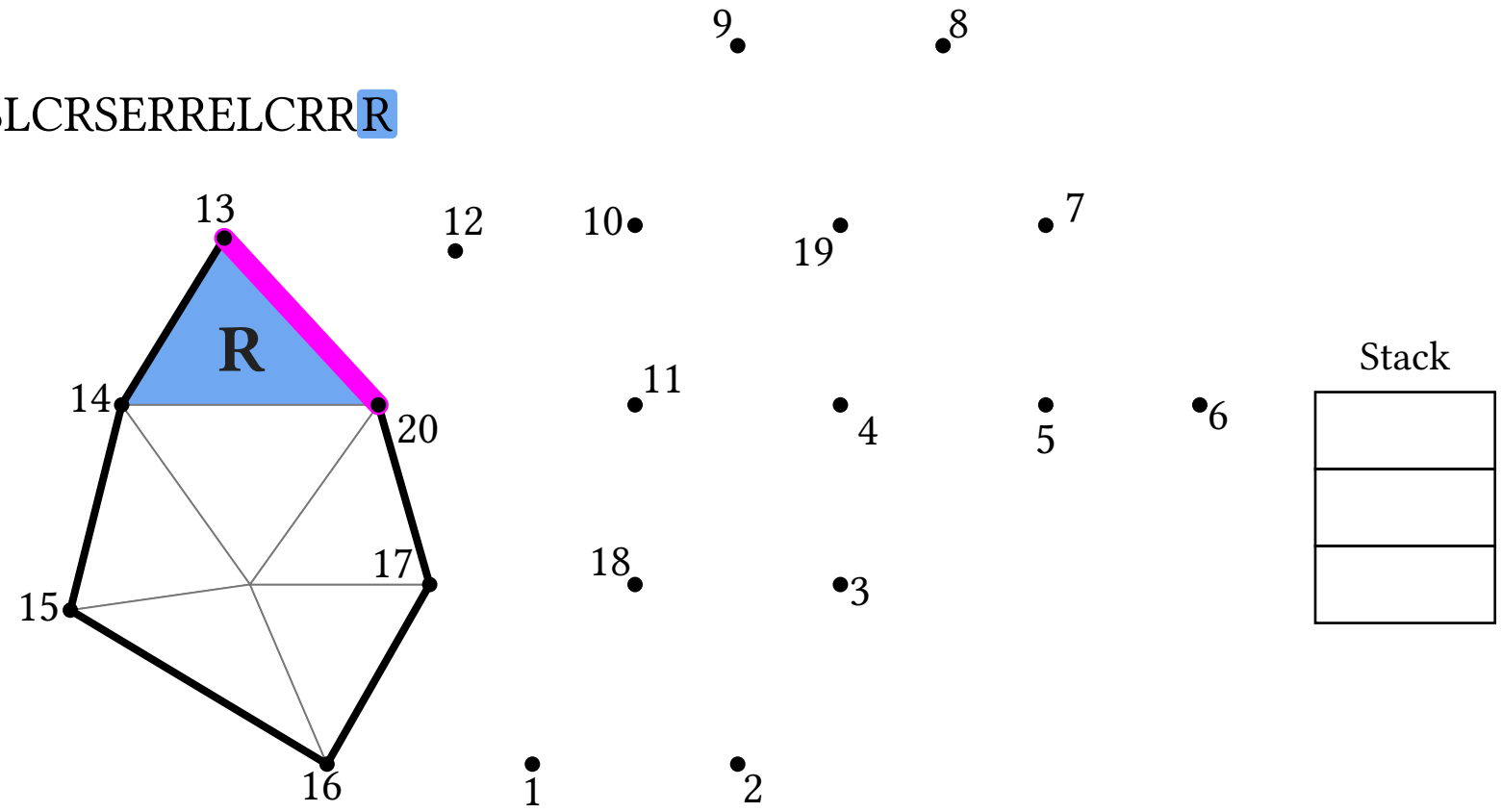


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRC

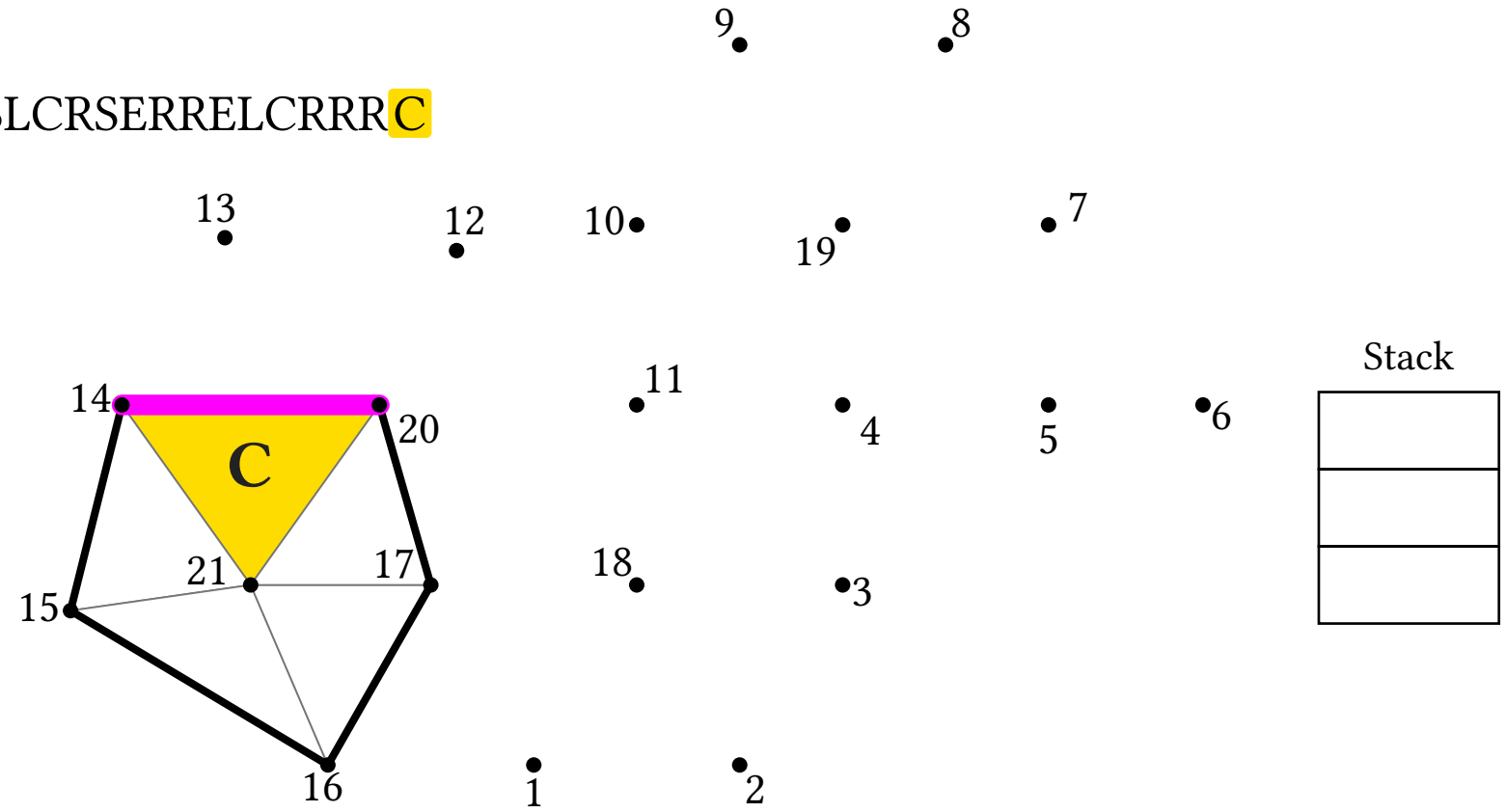


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCR

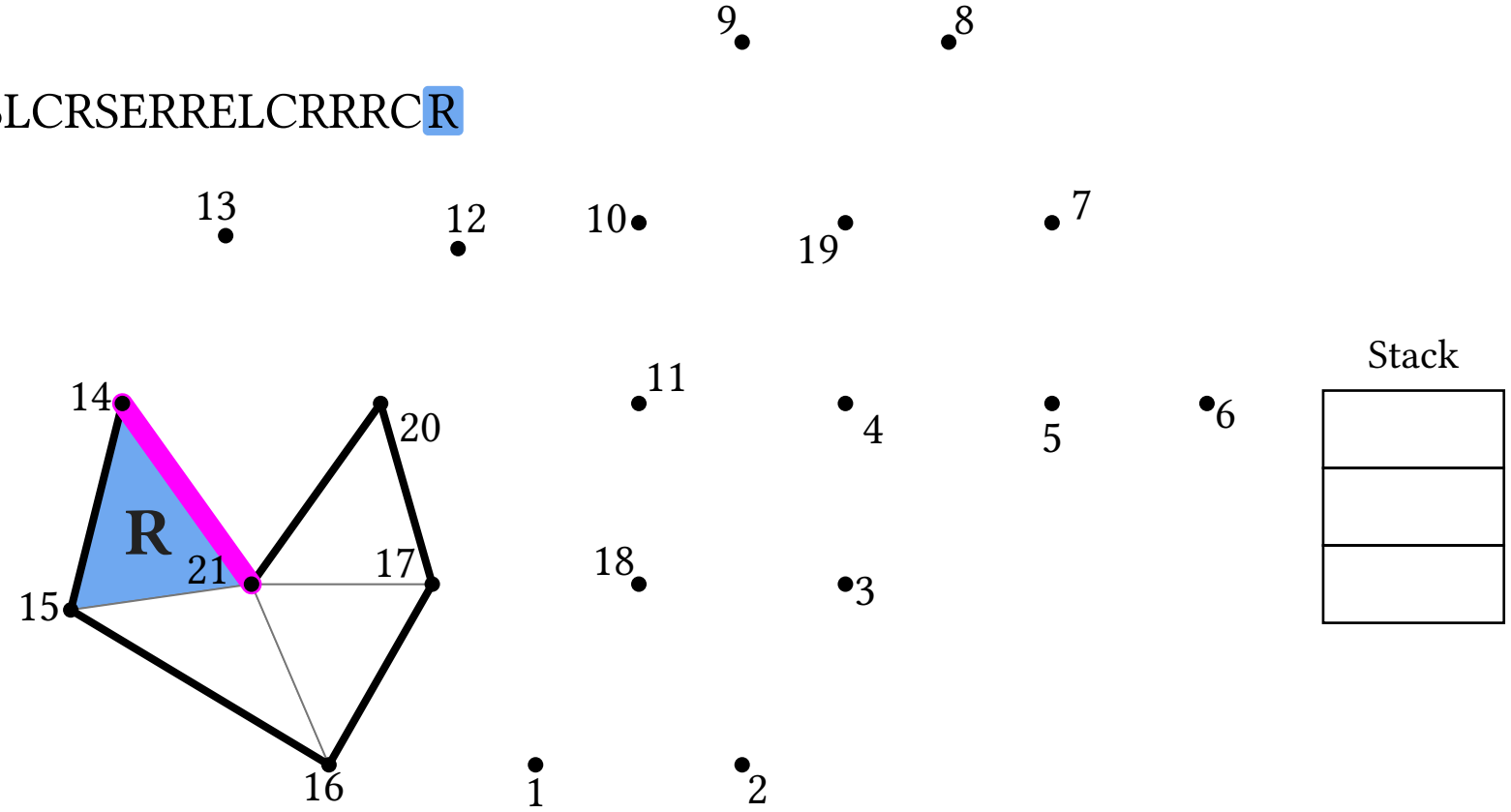


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRR

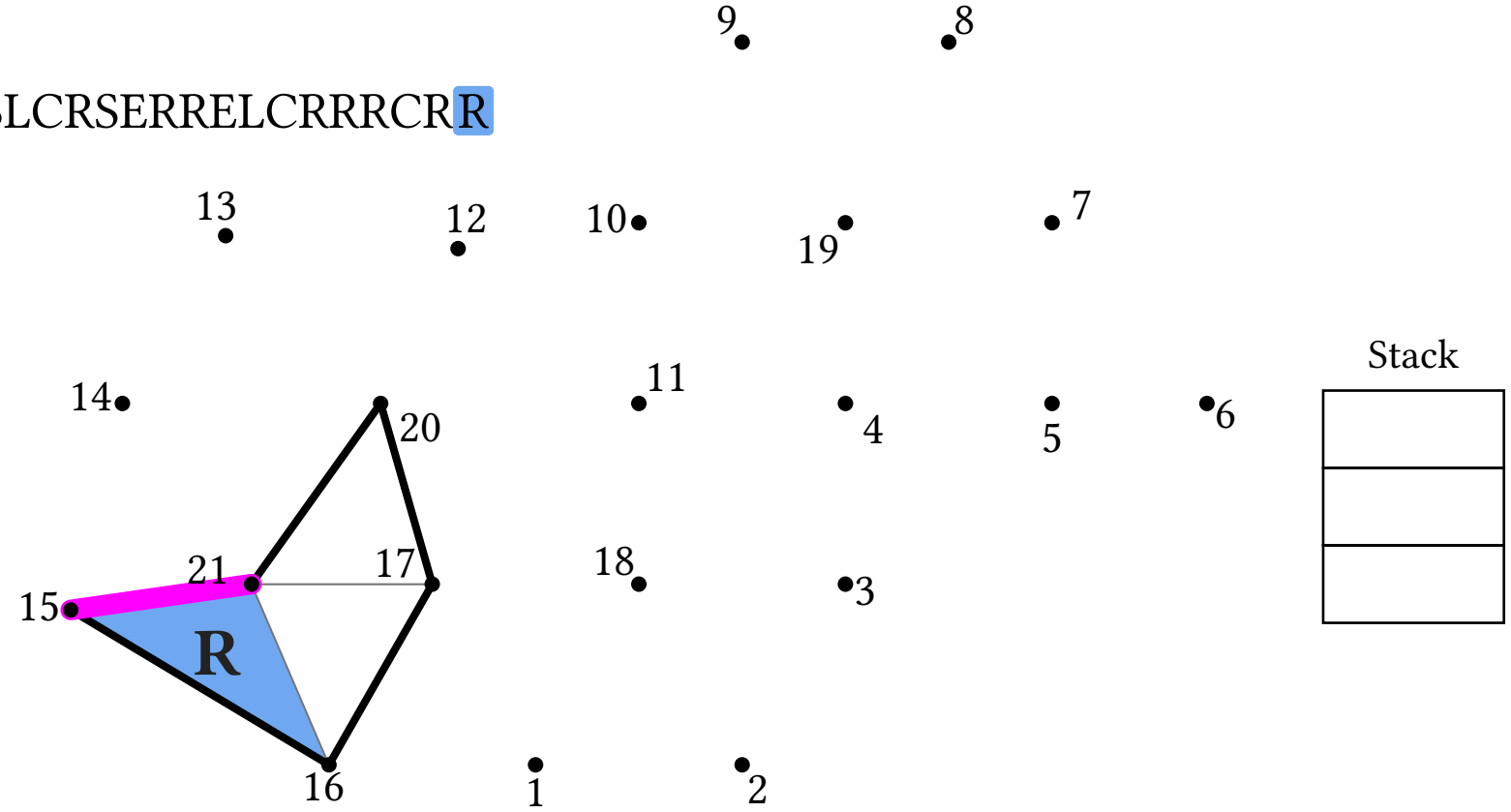


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRR

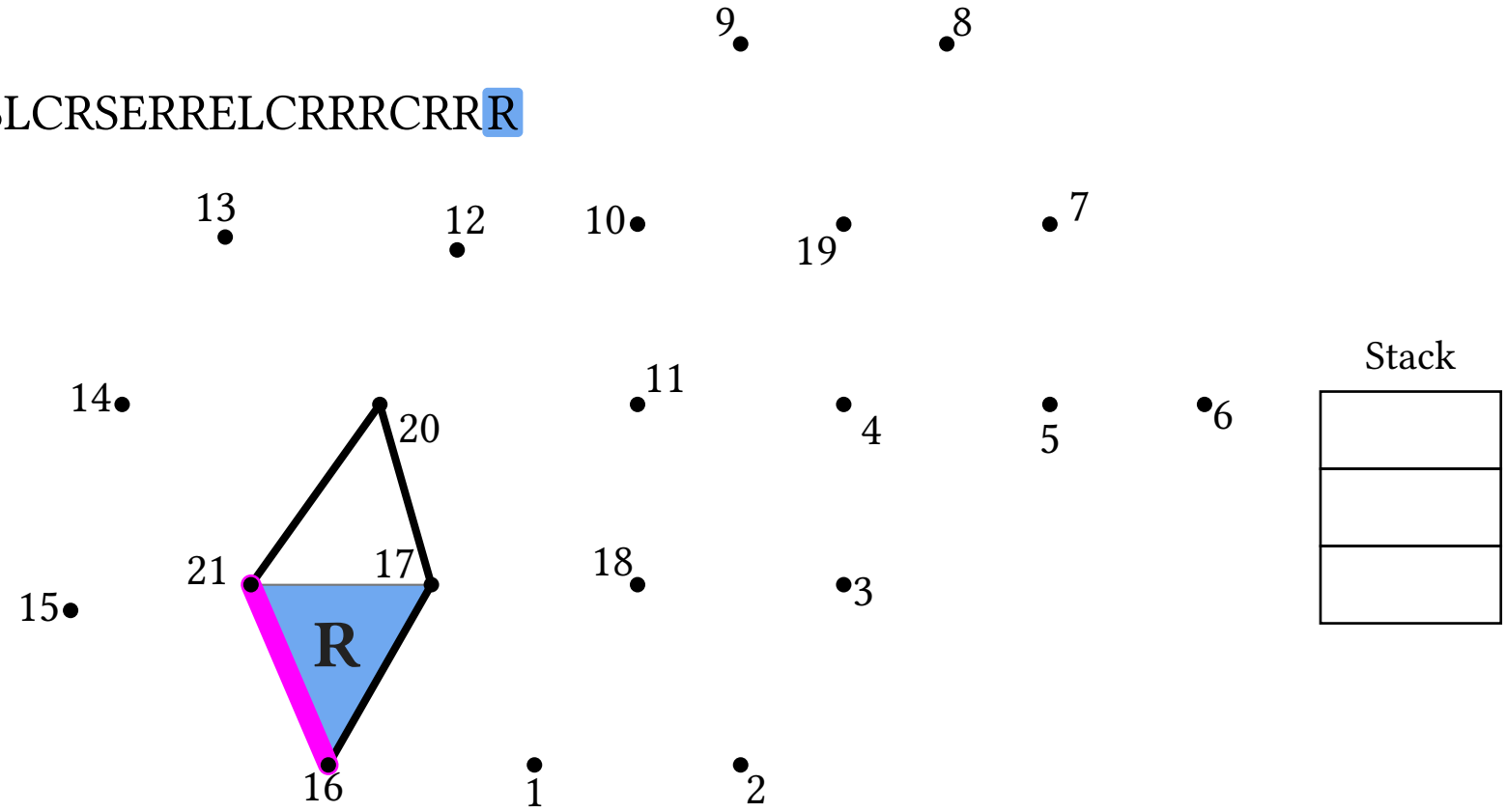


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

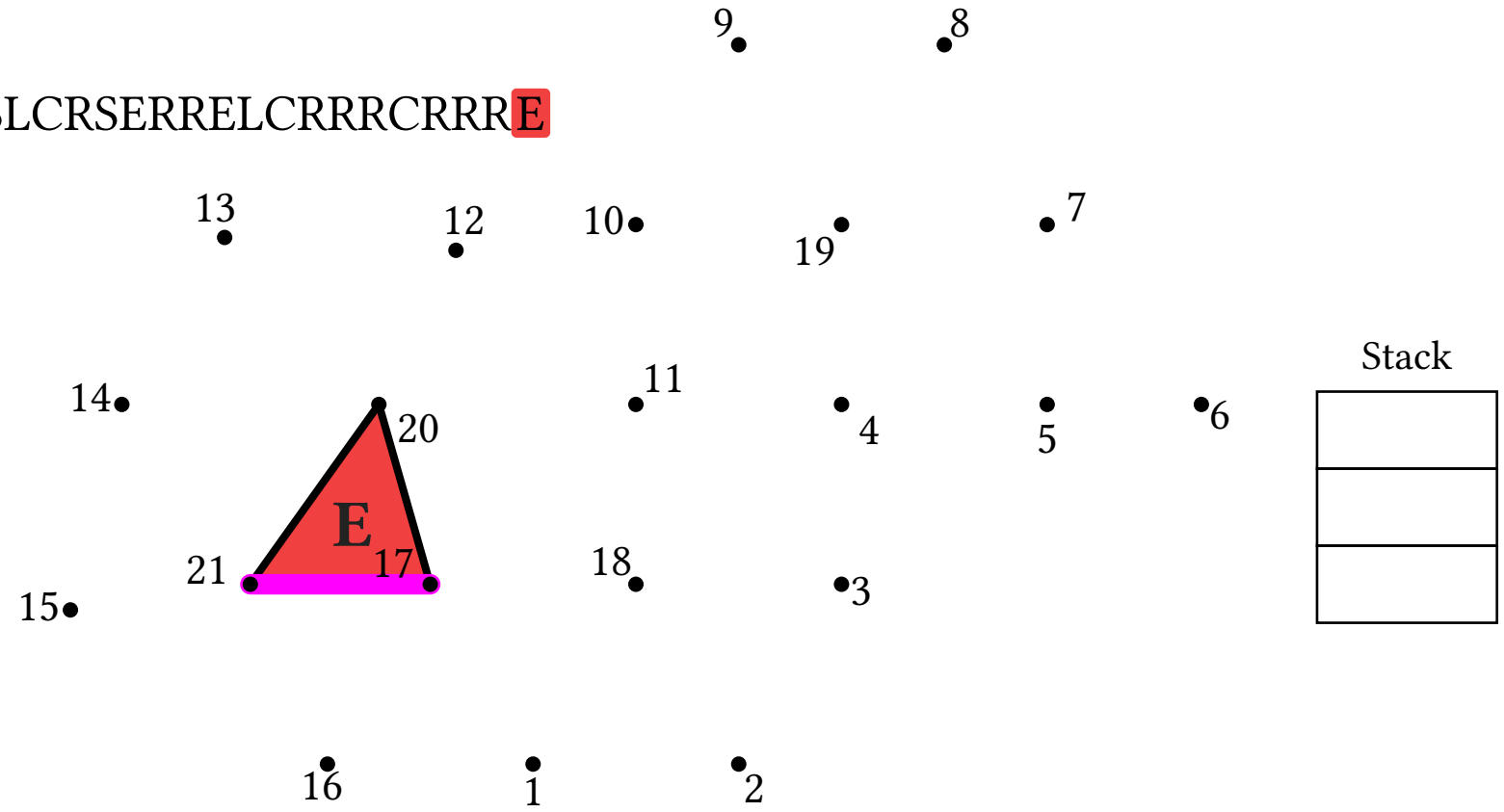


Figure 3: Jednoduchá komprese.

History = CCRRRSLCRSERRELCRRRCRRRE

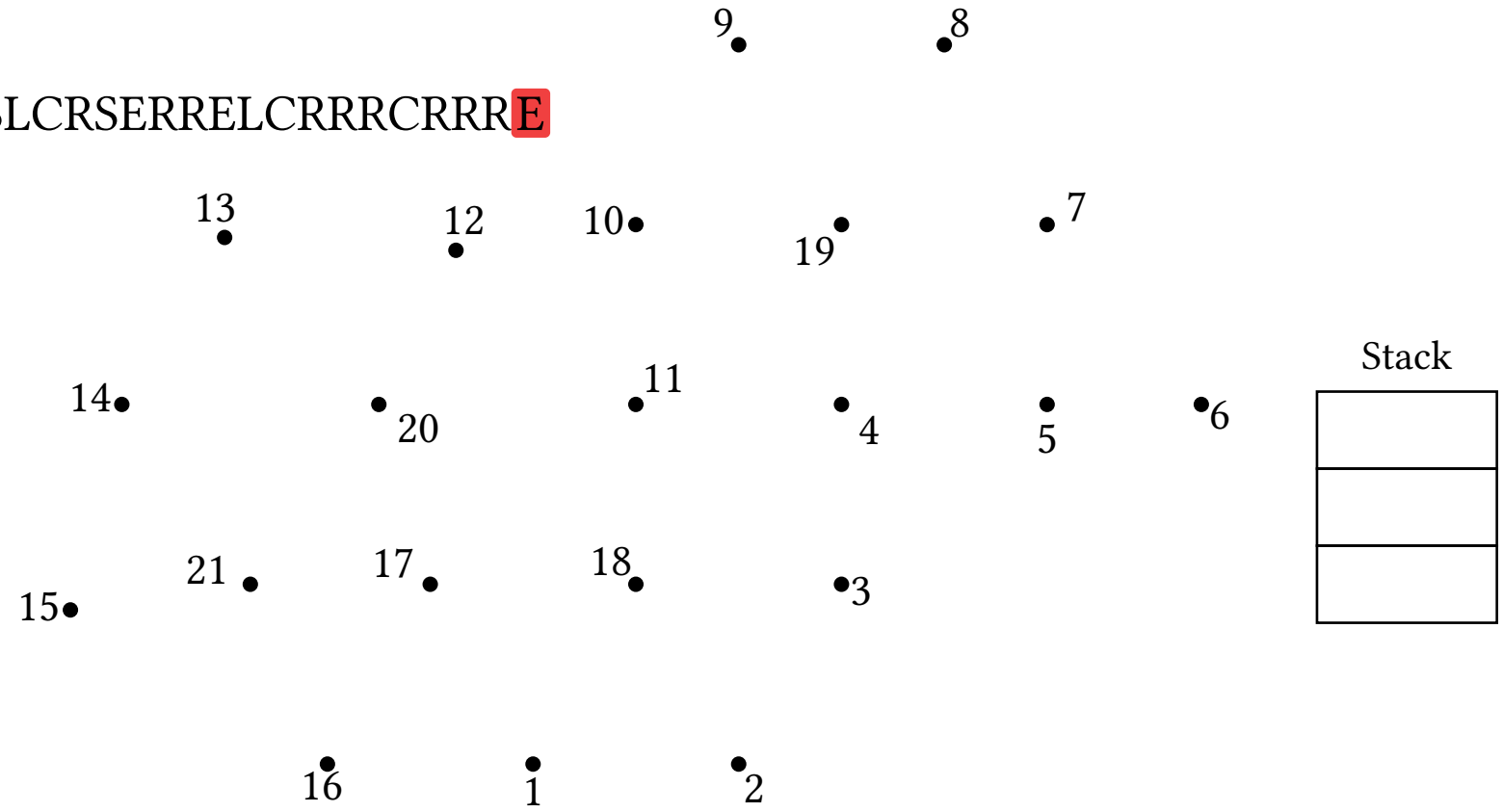


Figure 3: Jednoduchá komprese.

2.3 Vyšší genus?

History =

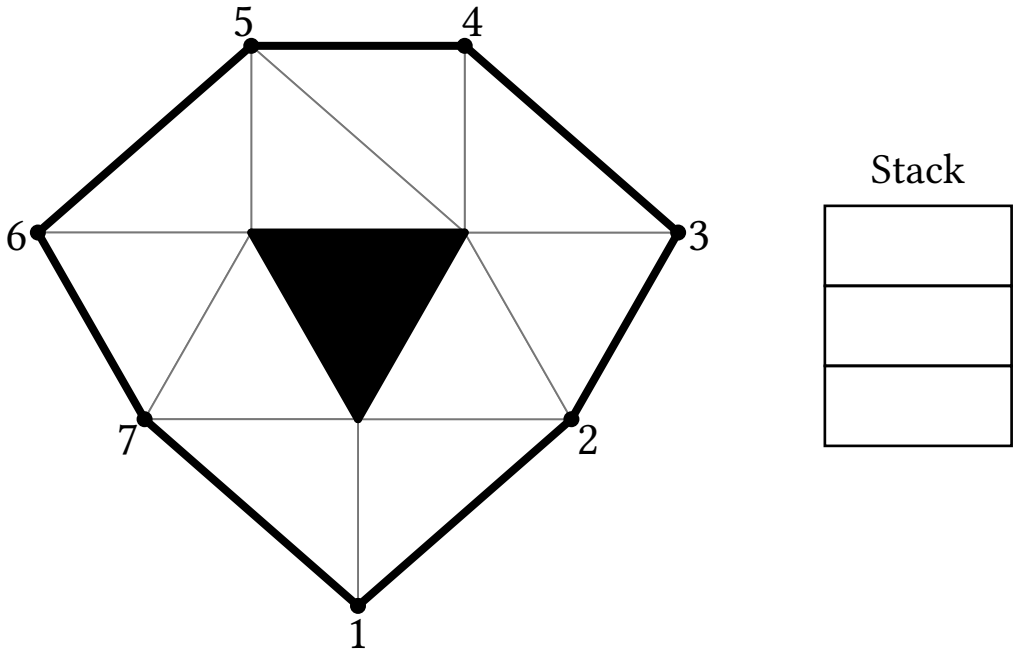


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = S

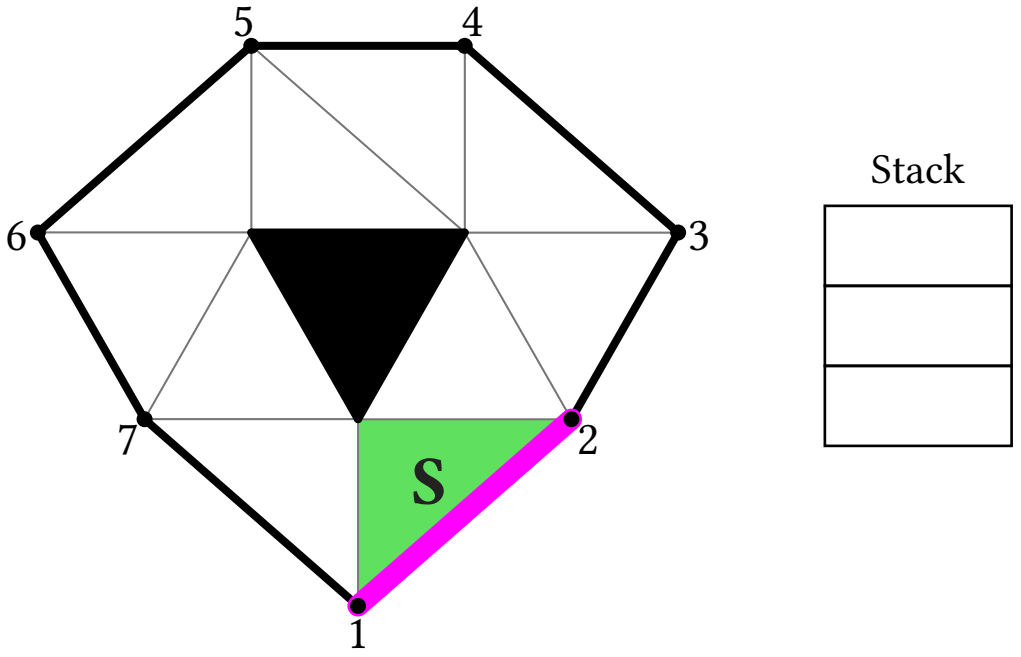


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = S

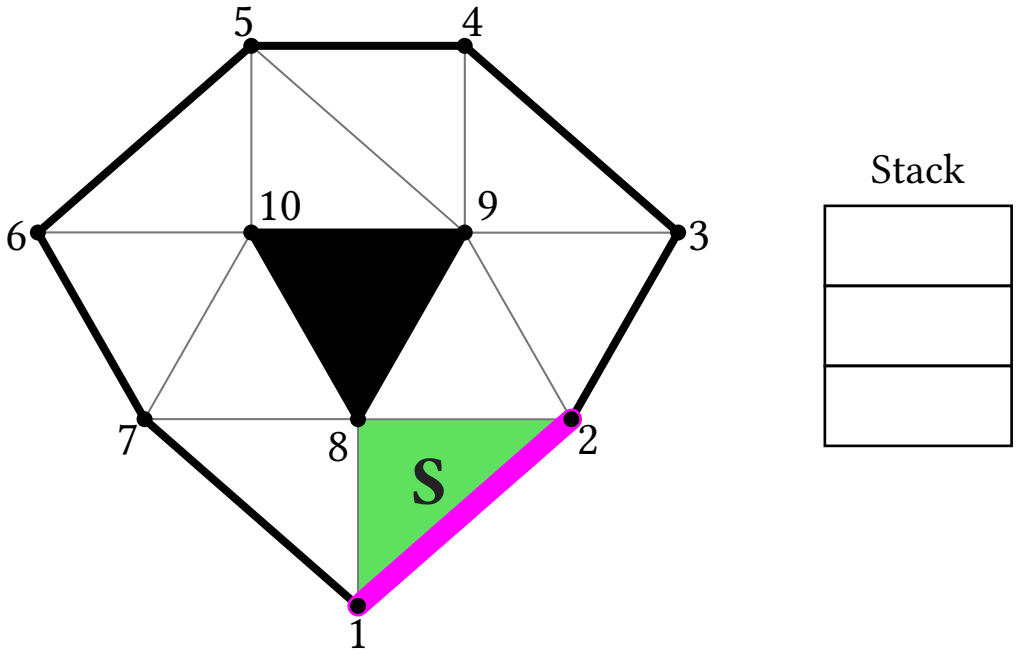


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SL

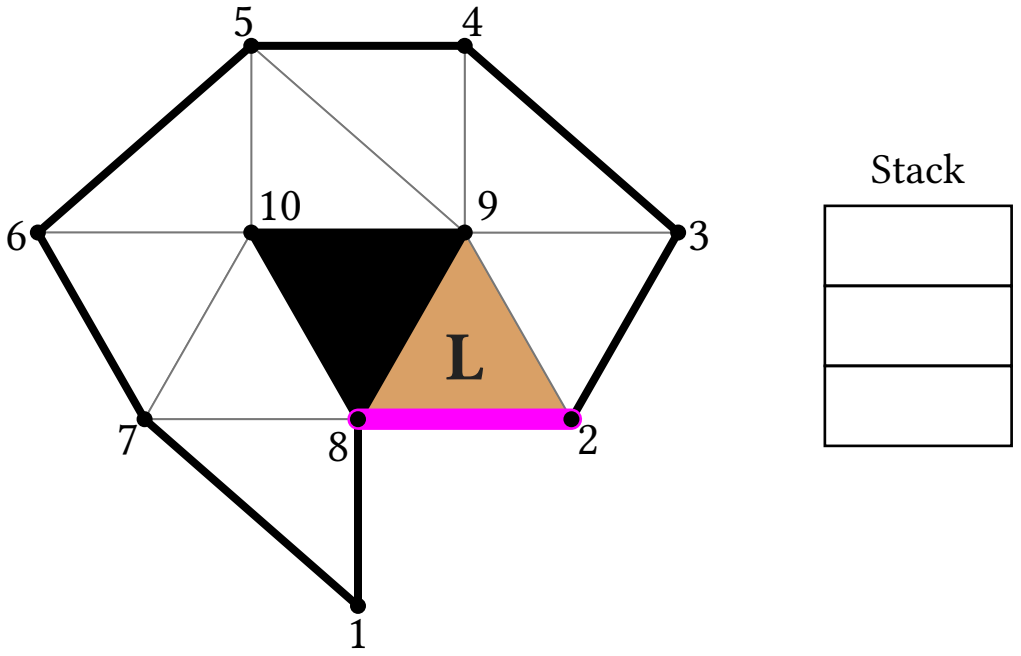


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLR

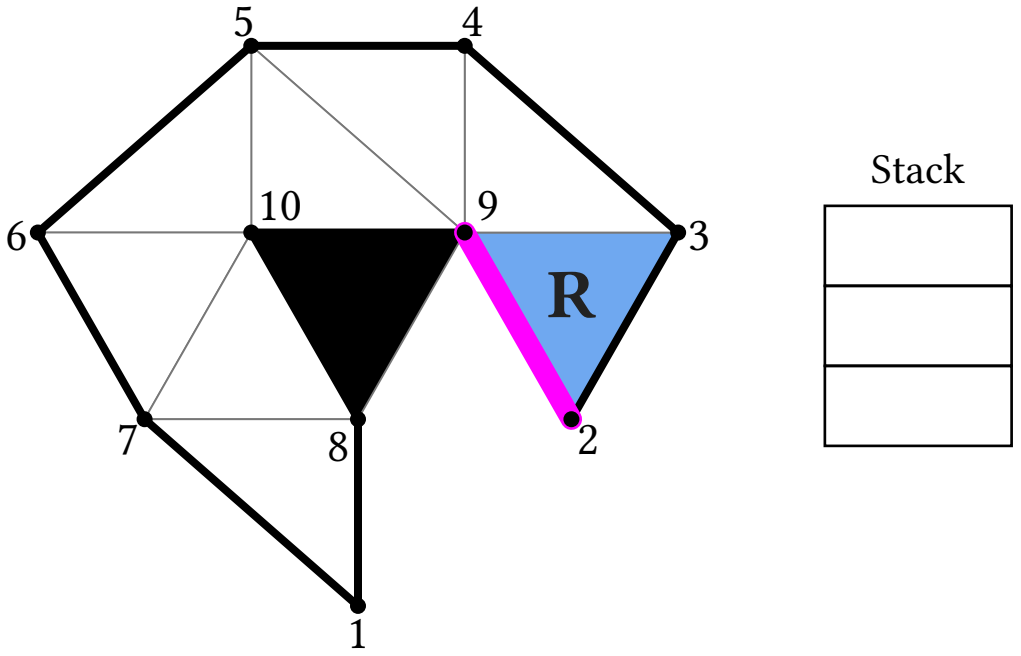


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRR

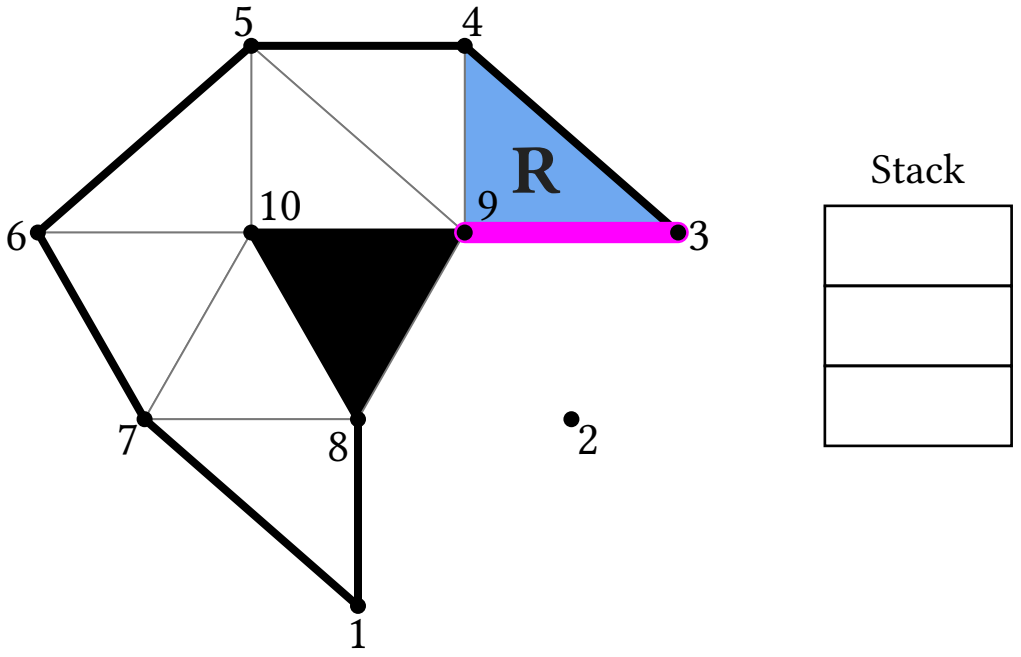


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRR

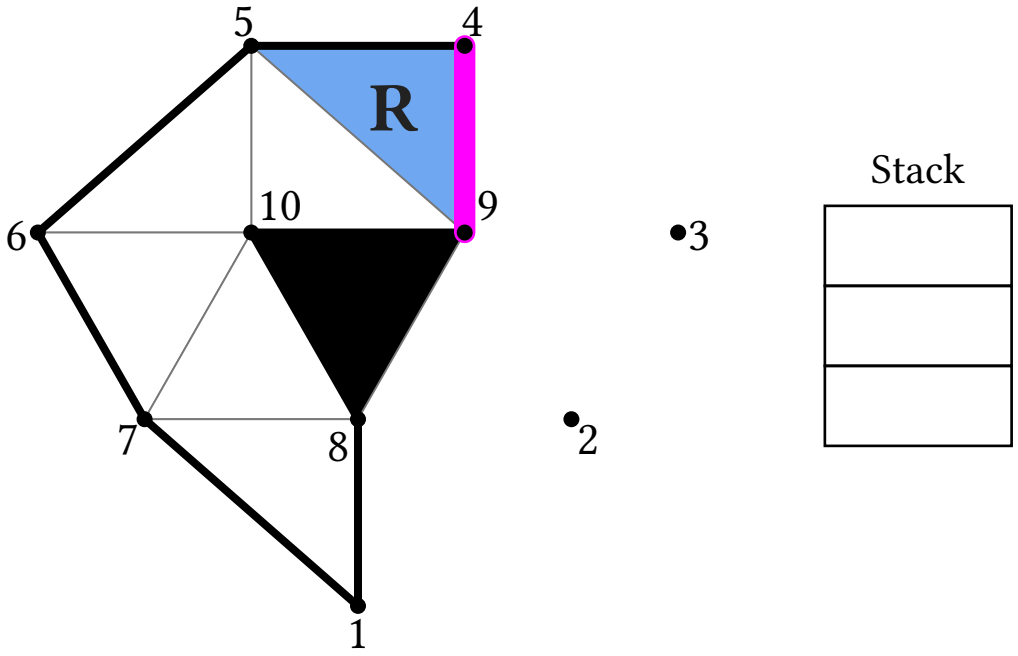


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRL

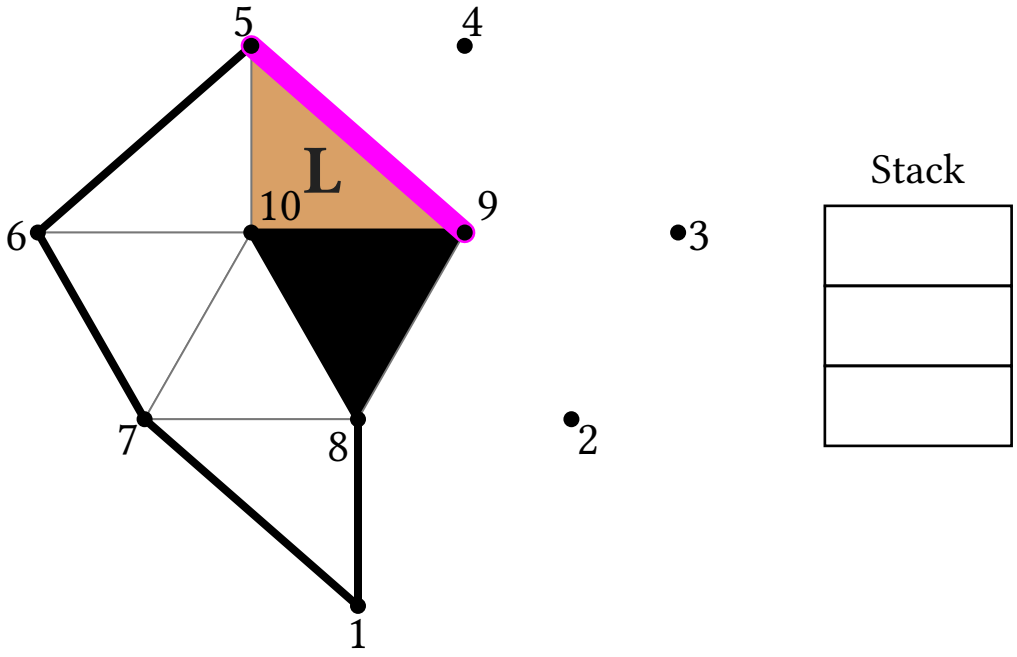


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLR

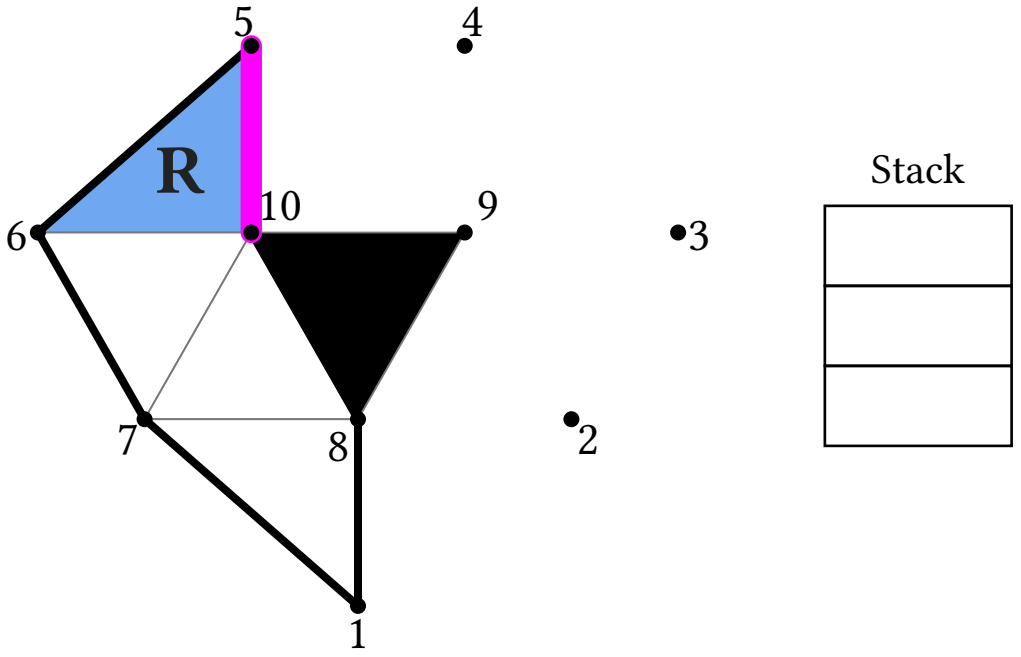


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRR

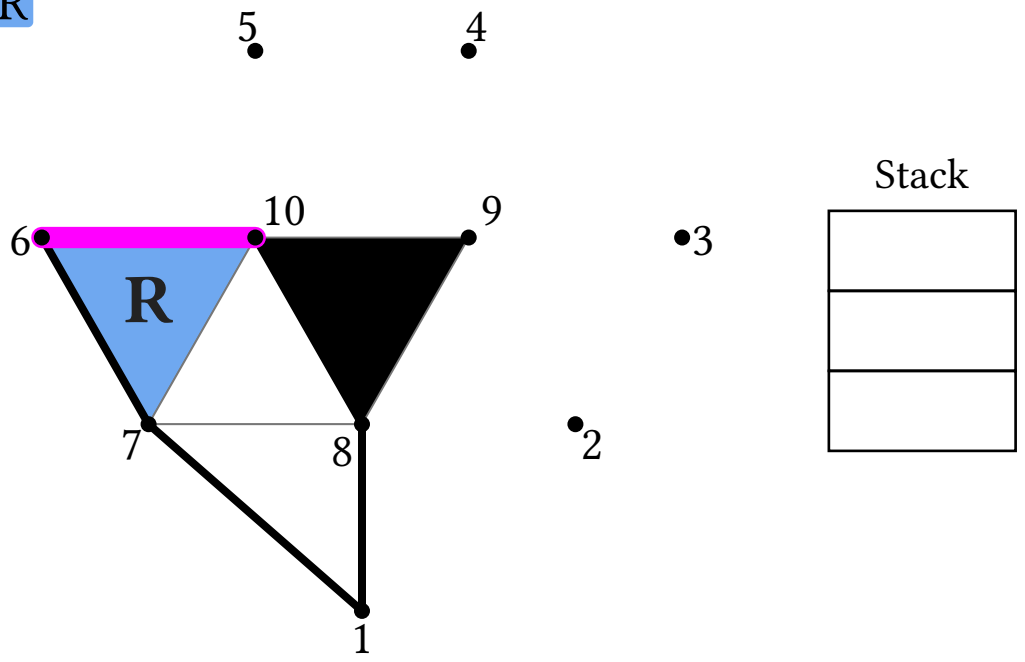


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRRL

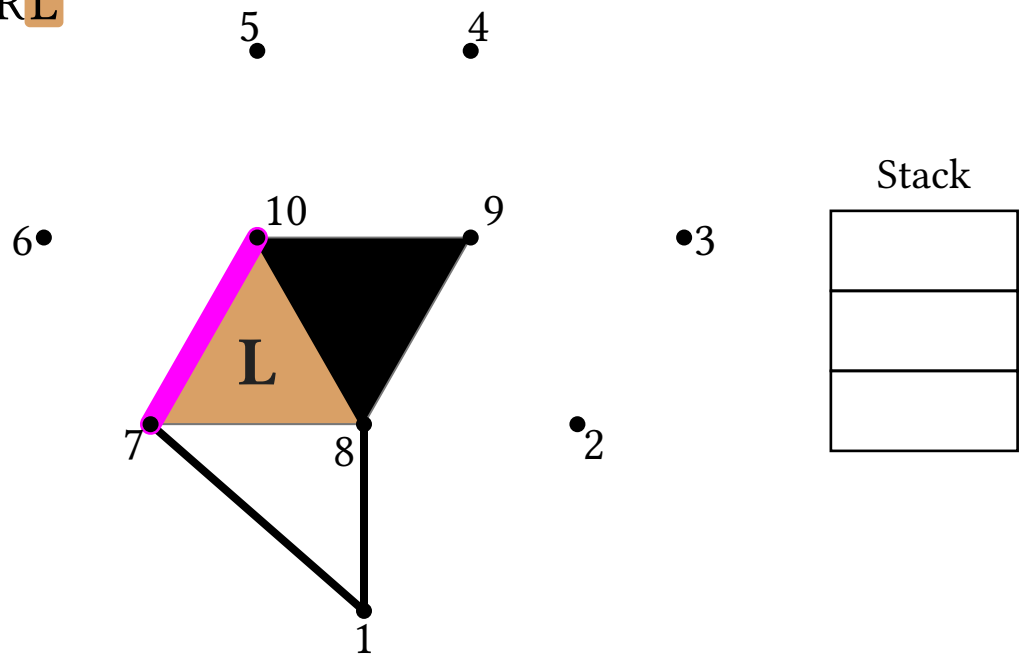


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRRLE

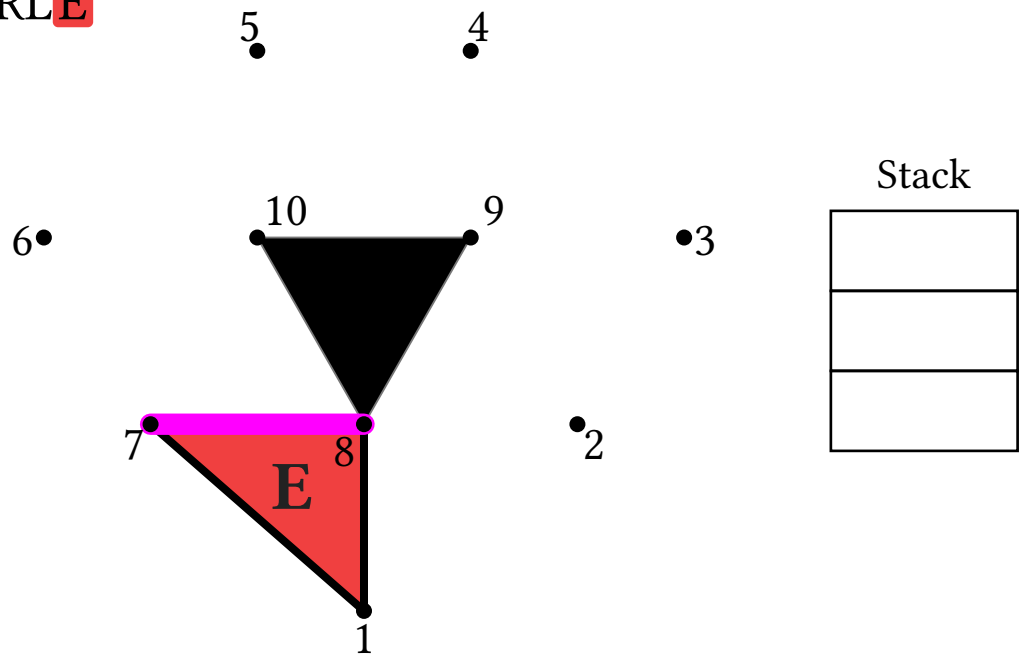


Figure 4: Komprese díry.

2.3 Vyšší genus?

History = SLRRRLRRLE

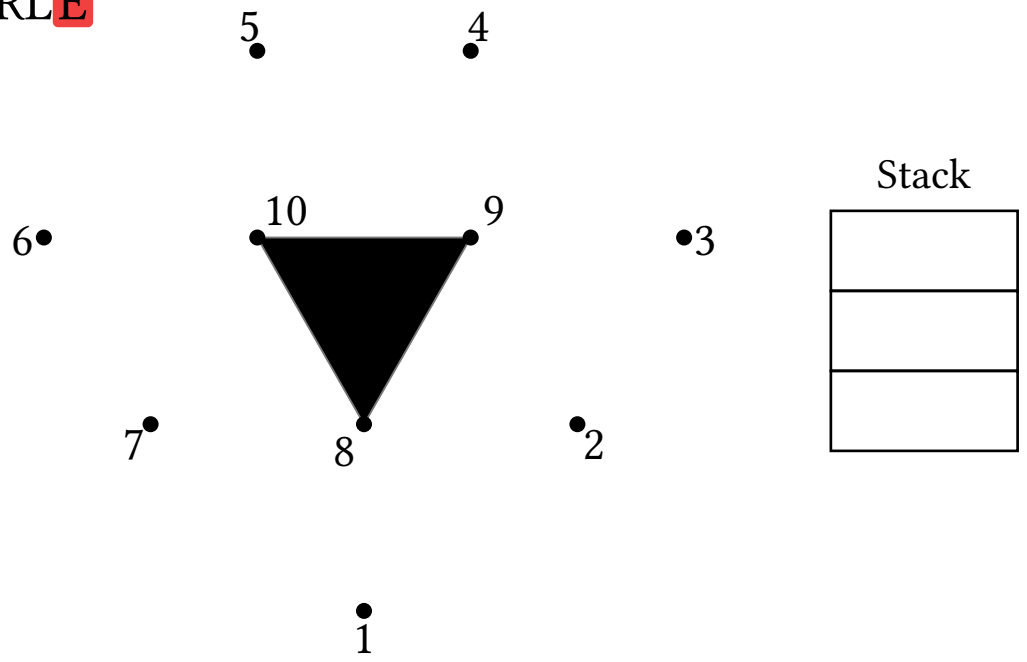


Figure 4: Komprese díry.

3. Dekomprese

3.1 Možnosti

3.2 Pravidla

History = CCRRRSLCRSERRELCRRRCRRRE

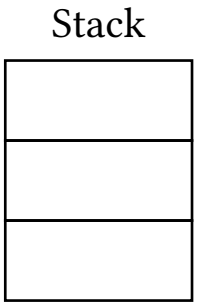


Figure 5: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE

Stack

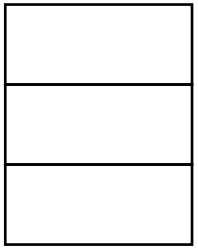
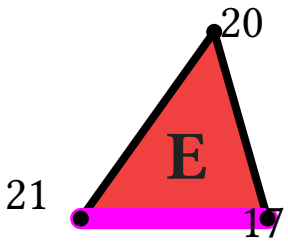


Figure 5: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRRE



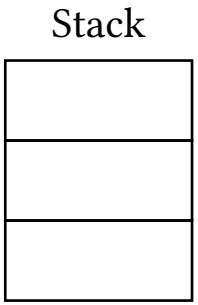
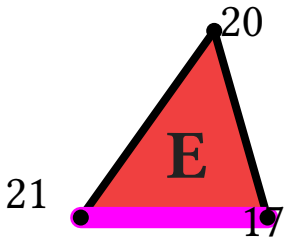


Figure 5: Dekomprese.

History = CCRRRSLCRSERRELCRRRCRRR



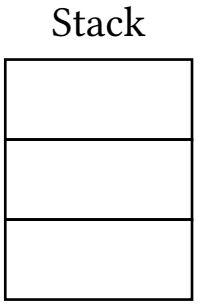


Figure 5: Dekomprese.

4. Zdroje

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4.2 Obrázky (dobový kontext)

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3	**Windows 98** (5/1998)	SoundCloud (artwork)
4	**MrBeast** (5/1998)	YouTube (avatar kanálu)
5	**Google** (9/1998)	logodix.com
6	**Half-Life** (11/1998)	Valve, valvearchive.com Wikipedia *fair use*
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8	**Futurama** (3/1999)	ČSFD (plakát)
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10	**Napster** (6/1999)	Jusore, GPLv3, Wikimedia Commons
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4.3 Reference

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